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Course Project 2 - Flexible Robotic Link

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Contents

1 Task 0: Problem Setup and Discretization **5**
- 1.1 Discretization Methods . . . 6
- 1.2 Analytic Jacobian Computation . . . 6

2 Task 1: Equilibria and Terminal Cost **7**
- 2.1 Equilibrium Manifold . . . 7
- 2.2 Cost Function . . . 8
- 2.3 Terminal Cost via DARE . . . 8
- 2.4 Step Size Selection (Armijo Rule) . . . 8

3 Task 2: Trajectory Generation (Newton's Method) **10**
- 3.1 Reference Generation . . . 10
  - 3.1.1 Task Definitions . . . 11
- 3.2 Performance Analysis: Stepwise vs. Polynomial Initialization . . . 13
  - 3.2.1 Limitations of Stepwise Initialization . . . 13
  - 3.2.2 Polynomial Interpolation and Dynamic Feasibility . . . 14
- 3.3 Optimization Challenges . . . 16
  - 3.3.1 Visual Analysis of the Local Minimum Trap . . . 17
- 3.4 Proposed Solution: Dynamic Horizon Extension . . . 18
  - 3.4.1 Theoretical Justification . . . 19
  - 3.4.2 Algorithm Implementation . . . 19
  - 3.4.3 Numerical Analysis and Validation . . . 19
  - 3.4.4 Convergence Analysis . . . 23

4 Task 3: Trajectory Tracking via LQR **25**
- 4.1 Linear Time-Varying (LTV) LQR . . . 25
- 4.2 Implementation Details . . . 25
- 4.3 Simulation Results . . . 26

5 Task 4: Trajectory Tracking via MPC **34**
- 5.1 MPC Formulation . . . 34
- 5.2 Key Features . . . 34
- 5.3 Simulation Results . . . 34

6 Task 5: Animation **43**
- 6.1 Visualization . . . 43

7 Conclusion **44**

8 Appendix **45**
- 8.1 Models matrices and parameters . . . 45
- 8.2 Analytical Derivation of the Jacobian . . . 45
  - 8.2.1 State-Space Formulation . . . 45
  - 8.2.2 Block A_{22}: Partial Derivative w.r.t. Velocities . . . 46
  - 8.2.3 Block A_{21}: Partial Derivative w.r.t. Positions . . . 46
  - 8.2.4 Jacobian Matrix B . . . 46
  - 8.2.5 Explicit Derivative Matrices . . . 46
- 8.3 Derivation of Discrete-Time Jacobians . . . 47
  - 8.3.1 Forward Euler Discretization . . . 47
  - 8.3.2 Runge-Kutta 4 (RK4) Discretization . . . 47
- 8.4 Newton Method with Dynamic Horizon Extension algorithm . . . 49

List of Figures

1 Double Pendulum System with flexible link. 5
2 3D Equilibrium manifold. 7
3 Equilibrium manifold as a function of θ_1. 7
4 Armijo line search behavior A. 9
5 Armijo line search behavior B. 9
6 Armijo line search behavior C. 9
7 Armijo line search behavior D. 9
8 Comparison of reference generation strategies. 10
9 Elbow Up 11
10 Swing Up 12
11 Swing Down 12
12 Complex Maneuver 13
13 Visualization of the defined control tasks and their target equilibria. 13
14 Newton's method applied to the Elbow Up maneuver using Stepwise reference. Note the control discontinuity manifesting as an infeasible torque spike. 14
15 Optimization results: Poly 5 reference applied to the Swing Up task. 14
16 Optimization results: Poly 5 reference applied to the Elbow Up task. 15
17 Optimization results: Poly 5 reference applied to the Swing Down task. 15
18 Illustration of the "Local Minimum Trap" where the solver converges to a suboptimal trajectory rather than the target equilibrium. 17
19 Cost function evolution during failure. The cost converges to a steady value, concealing the failure to reach the target. 18
20 Descent direction norm. The algorithm mathematically converges (norm $\to 0$), but to an incorrect solution. 18
21 Evolution of the trajectory during the Dynamic Horizon Extension process. The system incrementally "learns" the complex maneuver. 20
22 Step-wise evolution of the cost function. Each "step" corresponds to a horizon extension. 21
23 Norm of the descent direction. Spikes indicate the start of a new extension phase, followed by rapid convergence. 21
24 Optimization results: Poly 5 reference applied to the Swing Up task. 22
25 Optimization results: Poly 5 reference applied to the Elbow Up task. 23
26 Optimization results: Poly 5 reference applied to the Swing Down task. 23
27 Trajectory tracking on a swing-up maneuver 26
28 Error analysis during LQR tracking on a swing-up maneuver 27
29 Trajectory tracking on a swing down maneuver 28
30 Error analysis during LQR tracking of a swing down maneuver 29
31 Trajectory tracking on a elbow up maneuver 30
32 Error analysis during LQR tracking of elbow up maneuver 31
33 Trajectory tracking on a complex maneuver 32
34 Error analysis during LQR tracking of complex maneuver 33
35 Trajectory tracking on a swing up maneuver 35
36 Error analysis during MPC tracking of swing up maneuver 36
37 Trajectory tracking on a swing down maneuver 37
38 Error analysis during MPC tracking on a swing down maneuver 38
39 Trajectory tracking on a elbow up maneuver 39
40 Error analysis during MPC tracking of a elbow up maneuver 40
41 Trajectory tracking on a complex maneuver 41
42 Error analysis during MPC tracking on a complex maneuver 42

43 Double Pendulum Trajectories. Click any image to play the corresponding animation. 43

Abstract

Control of under-actuated robotic mechanisms with nonlinear joint stiffness remains a significant challenge due to the complex coupling between passive and active dynamics. This work develops an integrated framework for trajectory generation and tracking control for a planar two-degree-of-freedom double pendulum with nonlinear flexibility.

We first model the system dynamics and analyze the equilibrium manifold to identify feasible operating regions. For motion planning, we start with a Newton method for trajectory generation of simple to intermediate complexity maneuvers. To handle more complex trajectories that traverse challenging state-space regions, we introduce a Dynamic Horizon Extension alongside a Newton-based approach.

For tracking, we conduct a comparative analysis between a Linear Time-Varying Linear Quadratic Regulator (LTV-LQR) and Model Predictive Control (MPC). Results demonstrate that MPC outperforms LQR, specifically in handling constraints on the input, while LQR offers superior computational speed. The proposed framework successfully stabilizes the system along highly dynamic trajectories.

1 Task 0: Problem Setup and Discretization

The double pendulum model used in this project is reported in Figure 1.

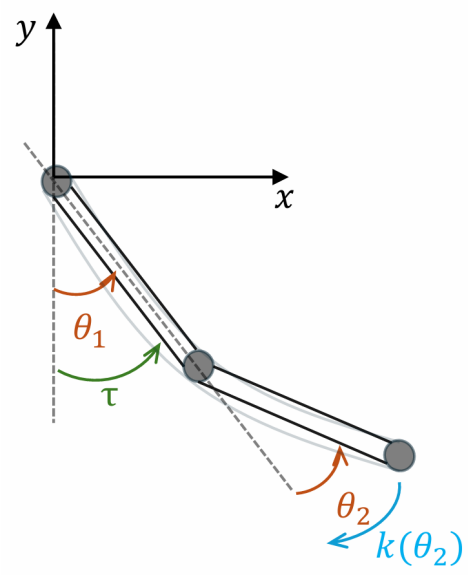


Figure 1: Double Pendulum System with flexible link.

The configuration vector is defined as: $q = \begin{bmatrix}\theta_1 & \theta_2\end{bmatrix}^\top \in \mathbb{R}^n$. With $n = 2$ degrees of freedom. The continuous-time dynamics of the double pendulum is governed by the equation:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + F\dot{q} + G(q) = \begin{bmatrix}\tau \\ 0\end{bmatrix} \tag{1}$$

Where[1]:

- $M(q) = M(q)^\top \succeq 0 \in \mathbb{R}^{n\times n}$ is the inertia matrix;
- $C(q,\dot{q}) \in \mathbb{R}^{n\times n}$ represents Coriolis and centrifugal forces;
- $F = F^\top \succ 0 \in \mathbb{R}^{n\times n}$ is the viscous friction matrix;
- $G(q)$ is the gravity vector;
- $\tau \in \mathbb{R}^{n_i}$ is the control torque applied at the first joint[2] with $n_i = 1$.

The adopted parameters for this system are summarized in Table 1. Some considerations regarding the model:

- **Under-actuation:** The system has two degrees of freedom (DoF) but only one control input (torque at the base);
- **Nonlinear Elasticity:** The stiffness between the links depends nonlinearly on the relative angle θ_2, complicating the dynamics compared to standard linear springs;
- **Low Friction:** The provided parameters (Set 2) involve relatively low friction coefficients, making the system prone to oscillations and sensitive to disturbances.

[1]The detailed expressions for $M(q)$, $C(q,\dot{q})$, F, and $G(q)$ can be found in the Appendix 8.1.
[2]In this tractation are not considered non-conservative forces or contact forces.

The state space vector is defined to include both positions and velocities: $x = \begin{bmatrix} \theta_1 & \theta_2 & \dot{\theta}_1 & \dot{\theta}_2 \end{bmatrix}^\top \in \mathbb{R}^{n_s}$. With $n_s = 2n = 4$ states. Thus, the continuous time state-space nonlinear dynamics can be expressed as:

$$\dot{x} = f(x, u) = \begin{bmatrix} \dot{q} & \ddot{q} \end{bmatrix}^\top = \begin{bmatrix} \dot{\theta}_1 & \dot{\theta}_2 & \ddot{\theta}_1 & \ddot{\theta}_2 \end{bmatrix}^\top. \tag{2}$$

Where:

- $u = \tau \in \mathbb{R}$ control input;
- $f : \mathbb{R}^{n_s} \times \mathbb{R} \to \mathbb{R}^{n_s}$ is a nonlinear function representing the system dynamics.

1.1 Discretization Methods

Two discretization schemes have been implemented in `solver.py`[3].

1. **Forward Euler (FE):**

$$x_{k+1} = x_k + \Delta t f(x_k, u_k) \tag{3}$$

   While simple, this method is numerically less accurate or unstable.

2. **Runge-Kutta 4th Order (RK4):**

$$x_{k+1} = x_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{4}$$

   This method maps all poles of the continuous-time system within the unit circle in the discrete-time domain, ensuring stability. However, its computation time is higher than the previous method.

Where:

- Δt is the discretization time step;
- x_k, u_k are the state and control input at time step k;
- k_1, k_2, k_3, k_4 are the RK4 intermediate slopes.

1.2 Analytic Jacobian Computation

For the Newton-based optimal control algorithm, accurate gradients of the dynamics are required. Relying on finite differences can introduce numerical noise, while symbolic differentiation (via `SymPy`) proved too slow for iterative solving.

Therefore, we implemented an **analytic Jacobian** computation in `model.py`.

- $A = \frac{\partial f(x,u)}{\partial x} = \nabla_1 f(x, u)^\top \in \mathbb{R}^{n_s \times n_s}$: State Jacobian;
- $B = \frac{\partial f(x,u)}{\partial u} = \nabla_2 f(x, u)^\top \in \mathbb{R}^{n_s \times n_i}$: Input Jacobian;

The Jacobians were derived manually and implemented using optimized NumPy operations (and Numba JIT compilation for speed). This approach ensures that the linearization required for the Backward Pass in Newton's method is both precise and computationally efficient. The detailed derivation is reported in Appendix 8.2.

[3]Details in Appendix 8.3

2 Task 1: Equilibria and Terminal Cost

2.1 Equilibrium Manifold

The system's equilibria are defined by states where $\dot{q} = \ddot{q} = 0$. This implies the gravity and stiffness torques must be balanced by the input τ. We analyzed the dynamics equation at equilibrium:

$$\cancel{M(\theta_1,\theta_2)\begin{pmatrix}\ddot{\theta}_1\\ \ddot{\theta}_2\end{pmatrix}} + \cancel{C(\theta_1,\theta_2,\dot{\theta}_1,\dot{\theta}_2)\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\end{pmatrix}} + \cancel{F\begin{pmatrix}\dot{\theta}_1\\ \dot{\theta}_2\end{pmatrix}} + G(\theta_1,\theta_2) = \begin{pmatrix}\tau\\ 0\end{pmatrix} \tag{5}$$

This reduces to:

$$\underbrace{\begin{pmatrix} gl_{c1}m_1\sin(\theta_1) + gm_2(l_1\sin(\theta_1) + l_{c2}\sin(\theta_1+\theta_2)) \\ gm_2l_{c2}\sin(\theta_1+\theta_2) + 3k\sin(\theta_2)\cos^2(\theta_2)\end{pmatrix}}_{G(\theta_1,\theta_2)} = \underbrace{\begin{pmatrix}\tau\\ 0\end{pmatrix}}_{\tau} \tag{6}$$

In this framework, we have three variables: θ_1, θ_2, and τ. By fixing one of the three variables, we can solve for the other two. It looks natural, since we are interested in obtaining specific configurations, and since the first joint is actuated, to fix θ_1 and solve for θ_2 and τ.

Using this approach, we generate an equilibrium manifold that is a function of the first joint angle θ_1. It is worth noting that two distinct configurations arise for each θ_1 value, corresponding to different θ_2 angles (e.g., "Elbow-Up" and "Elbow-Down" positions). They will respectively be stable and unstable configurations.

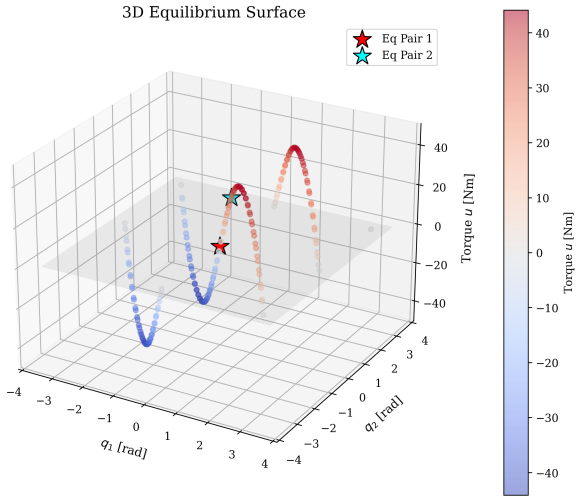


Figure 2: 3D Equilibrium manifold.

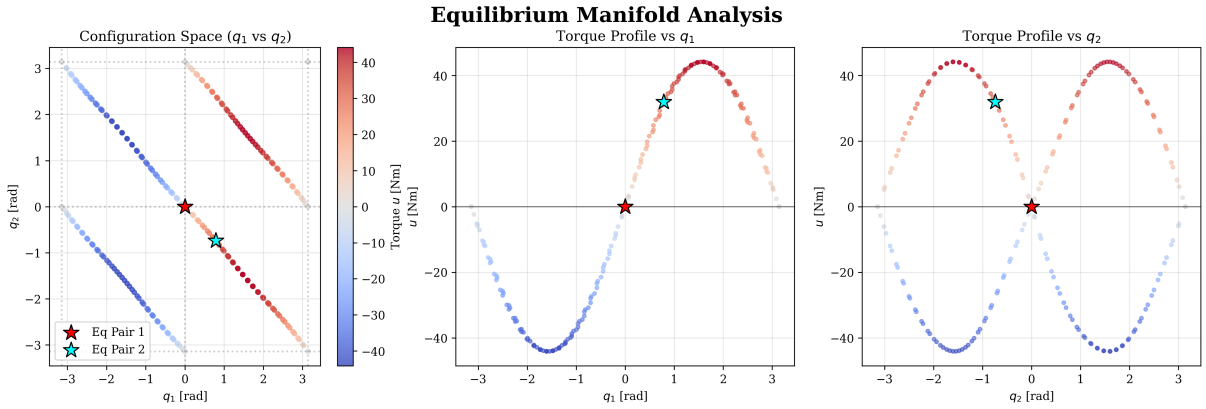


Figure 3: Equilibrium manifold as a function of θ_1.

The first thing we can see is that the manifold has a sinusoidal shape and a periodic behavior, which is expected given the rotational nature of the joints. From the plot, we can also observe some important characteristics:

- **Branches:** We can say that the manifold is **constituted by branches**, hence there will be starting and ending equilibria that are on the same branch and others that will be in two different ones.
  This will have an impact on the trajectory generation, as moving between different branches may require highly dynamical maneuvers since it's not possible to switch branch by steady-state maneuvers.
- **Uncontrolled second arm:** Since the second joint is unactuated, the torque τ is only applied at the first joint.
  This means that the system's ability to reach certain **configurations** in a given **time** is **limited by the dynamics of the second link** and the coupling between the two joints. This coupling will prove useful in achieving a stabilization of the second link by appropriately controlling the first one. We will see this is **not always possible**, and this causes problems in trajectory generation when trying to reach certain target equilibria and mantain them for some time.

2.2 Cost Function

For the optimal control problem, we defined the quadratic cost function:

$$J = \sum_{t=0}^{T-1} \left(\frac{1}{2} \Delta x_t^\top Q \Delta x_t + \frac{1}{2} \Delta u_t^\top R \Delta u_t \right) + \frac{1}{2} \Delta x_T^\top Q_T \Delta x_T \tag{7}$$

where $\Delta x = x - x_{ref}$ and $\Delta u = u - u_{ref}$. We selected $Q = \text{diag}(1, 1, 1, 1)$ and $R = 10^{-3} I$ to penalize state deviation while allowing for aggressive control actions.

2.3 Terminal Cost via DARE

A critical aspect of the finite-horizon optimization is the terminal cost Q_T. Simply setting $Q_T = Q$ often fails to stabilize the system at the end of the horizon, especially for short durations. In fact, without an appropriate terminal cost, the solver may produce trajectories that approach but do not settle at the desired equilibrium, leading to oscillations or drift post-optimization. It is as if the optimizer result was **"myopic"**, focusing only on minimizing cost up to time T without considering the long-term behavior beyond this point. To address this, we computed Q_T by solving the **Discrete Algebraic Riccati Equation** (DARE) based on the linearization at the final target equilibrium (x_{eq}, u_{eq}).

$$P_\infty = A^\top P_\infty A - (A^\top P_\infty B)(R + B^\top P_\infty B)^{-1}(B^\top P_\infty A) + Q \tag{8}$$

Setting $Q_T = P_\infty$ provides an approximation of the infinite-horizon cost-to-go. This essentially informs the solver that the trajectory continues optimally after time T, significantly improving convergence and stability for tasks like the "Swing Up".

2.4 Step Size Selection (Armijo Rule)

To ensure global convergence of the Newton method, we implemented a **line search using the Armijo rule**. The step size α is reduced ($\alpha \leftarrow \beta\alpha$) until the actual cost reduction is a sufficient fraction of the predicted reduction (based on the gradient). This prevents the solver from taking overly aggressive steps that could destabilize the trajectory and **ensures steady progress towards the optimum**. Additionally, the implementation includes a gradient check that validates the analytic descent direction by comparing it to a numeric slope $\frac{J(u+\epsilon\Delta u)-J(u)}{\epsilon}$ computed via finite difference. If the Armijo search fails to satisfy the sufficient decrease condition within the maximum allowed iterations, the **step size is set to zero** to prevent the update and safeguard the solver from adopting an unstable or divergent trajectory. Here there are some plots of the Armijo line search behavior during the trajectory optimization.

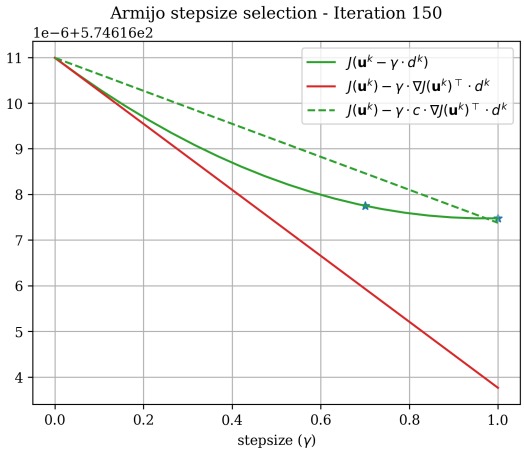


Figure 4: Armijo line search behavior A.

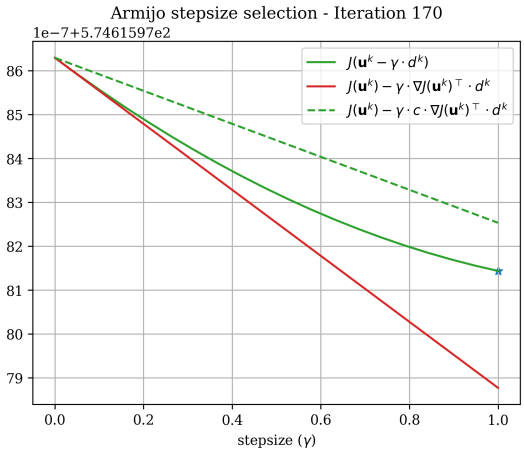


Figure 5: Armijo line search behavior B.

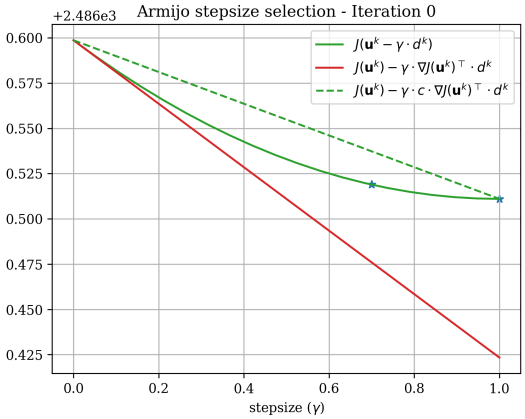


Figure 6: Armijo line search behavior C.

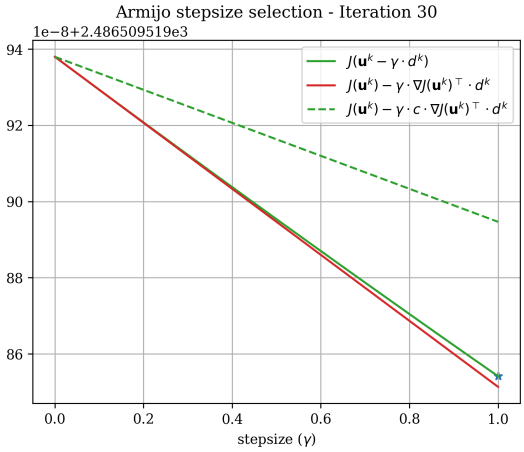


Figure 7: Armijo line search behavior D.

3 Task 2: Trajectory Generation (Newton's Method)

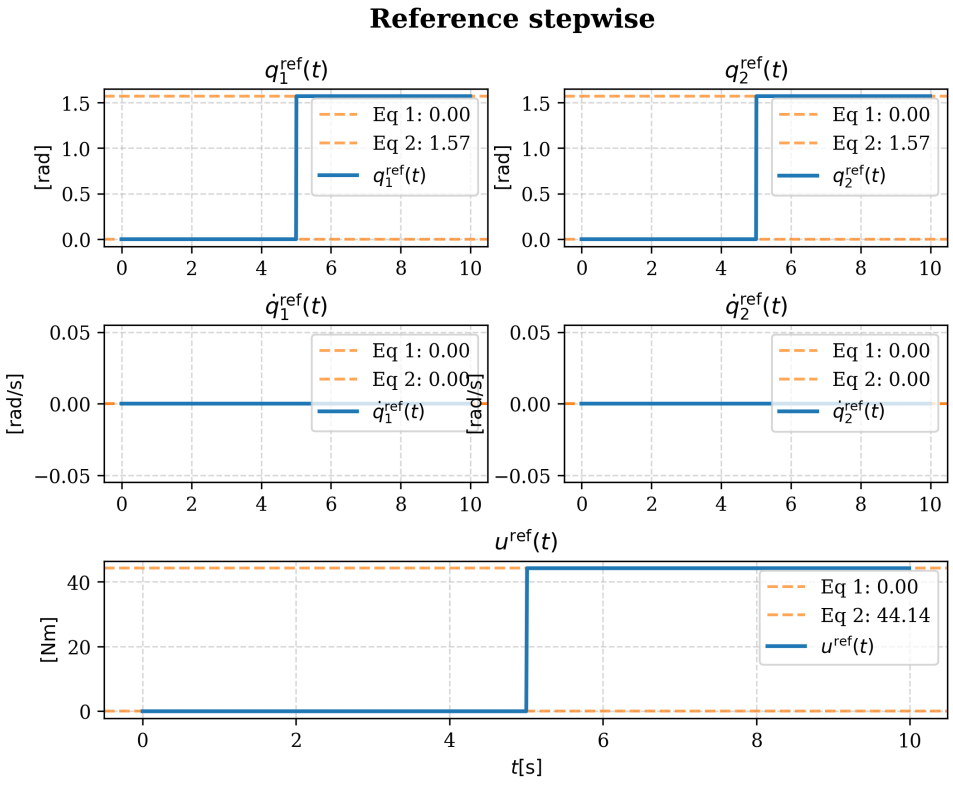
To generate feasible trajectories transitioning between system equilibria, we employed a Newton-type method for optimal control.

3.1 Reference Generation

Given the non-convex nature of the optimization landscape, the solver requires a reasonable reference trajectory to guide convergence, even if's dynamically infeasible. We analyzed two distinct generation methods:

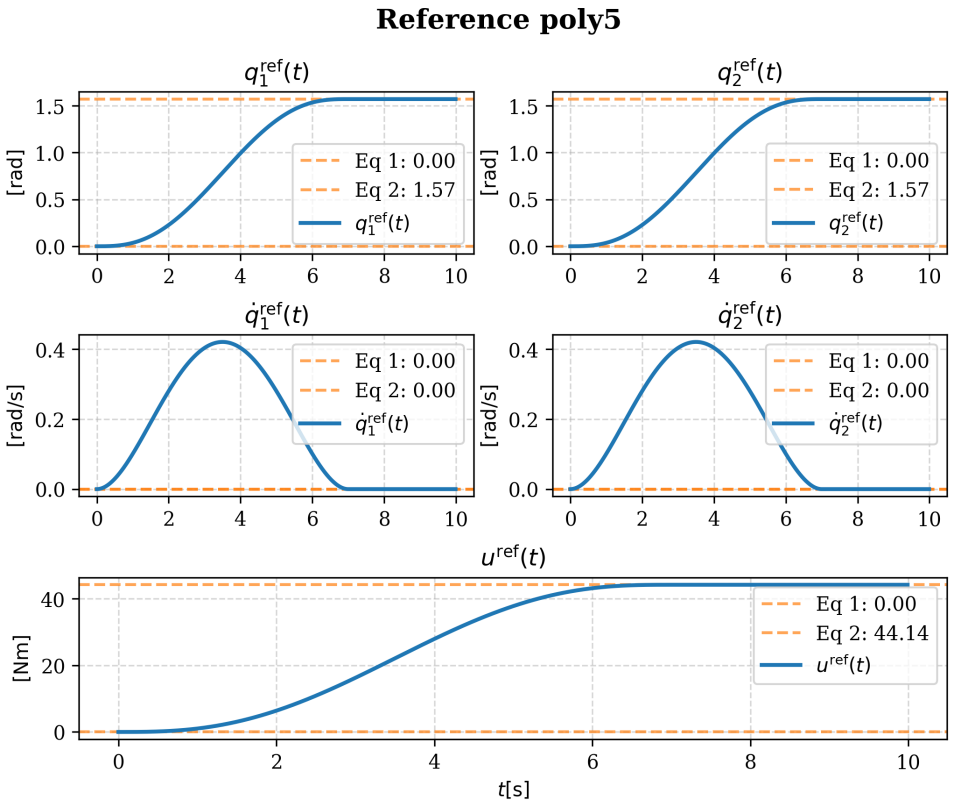
1. **Stepwise:** A naive approach involving an instantaneous switch from the initial state x_{start} to the goal state x_{goal}. This introduces a large discontinuity in the error term, often leading to poor convergence or aggressive control spikes.

2. **Polynomial Interpolation (Poly 5):** We generated smooth references for joint positions and velocities using 5th-order polynomials. This method ensures kinematic consistency by respecting boundary conditions (position, velocity), thereby significantly aiding the solver in finding a feasible solution.



(a) Stepwise Reference

Figure 8: Comparison of reference generation strategies.



(b) Poly 5 Reference

3.1.1 Task Definitions

We defined a set of standard maneuvers of varying complexity to evaluate the generation algorithms, ranging from simple transitions to complex swings.

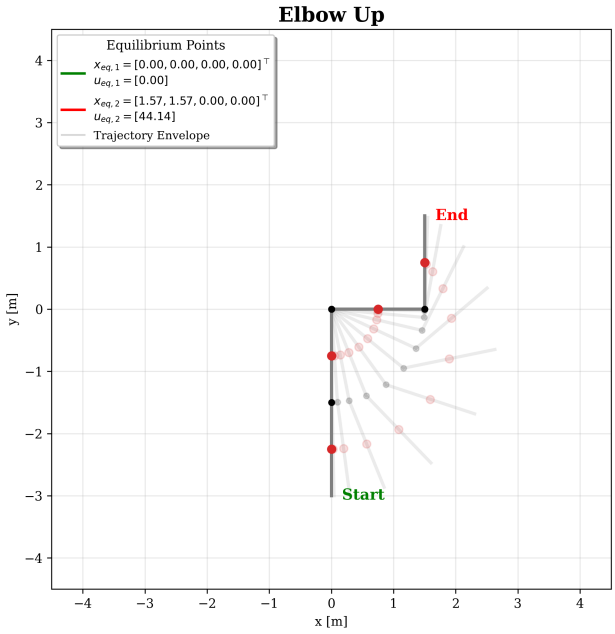


Figure 9: Elbow Up

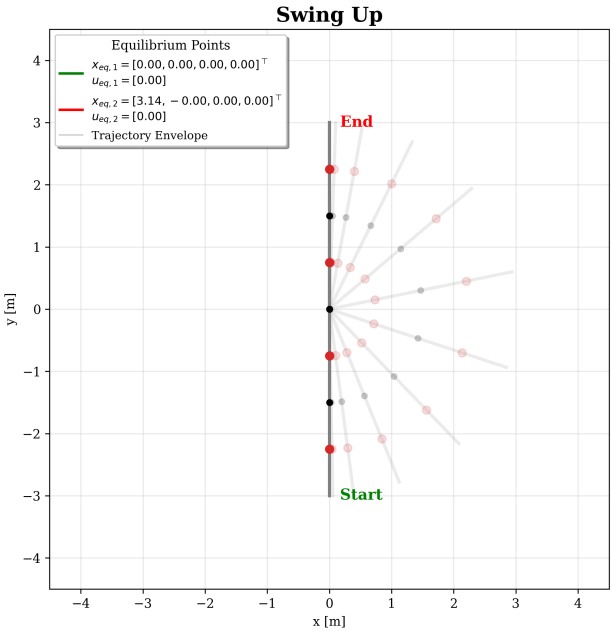


Figure 10: Swing Up

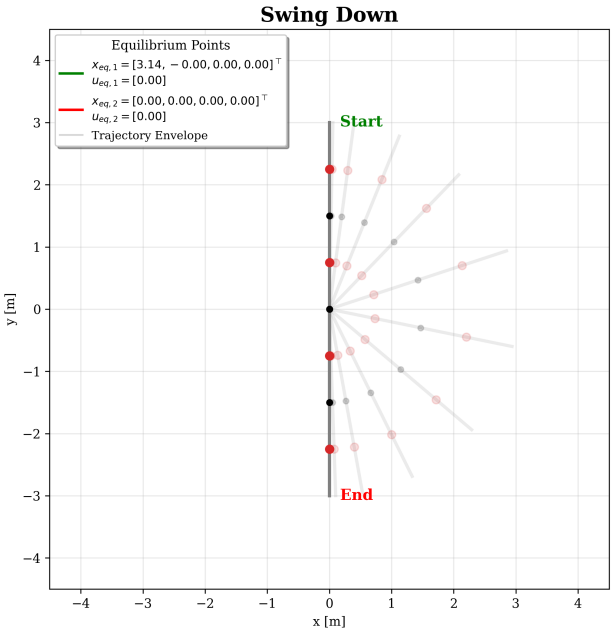


Figure 11: Swing Down

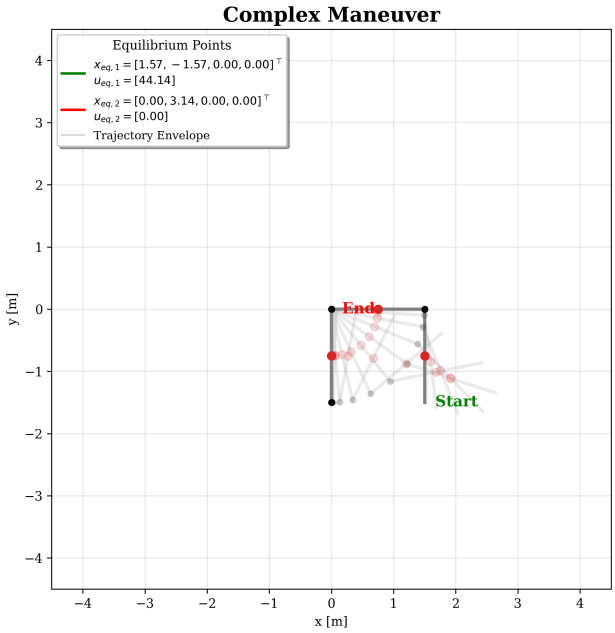


Figure 12: Complex Maneuver

Figure 13: Visualization of the defined control tasks and their target equilibria.

3.2 Performance Analysis: Stepwise vs. Polynomial Initialization

The convergence properties of Newton-type methods are highly sensitive to the quality of the initial reference trajectory. Our analysis indicates that the initialization strategy is a determinant factor in the solver's ability to locate a feasible solution within the under-actuated dynamic constraints.

3.2.1 Limitations of Stepwise Initialization

Applying Newton's method to the **Stepwise** reference resulted in significant numerical instability. As illustrated in Figure 14, the instantaneous state transition implies an infinite velocity and acceleration requirement at the discontinuity. This generates a large localized error gradient, causing the solver to propose impulsive control torques (u). Such spikes often drive the system state into regions of the manifold where the linearization approximation breaks down, leading to divergence or stabilization at infeasible equilibria.

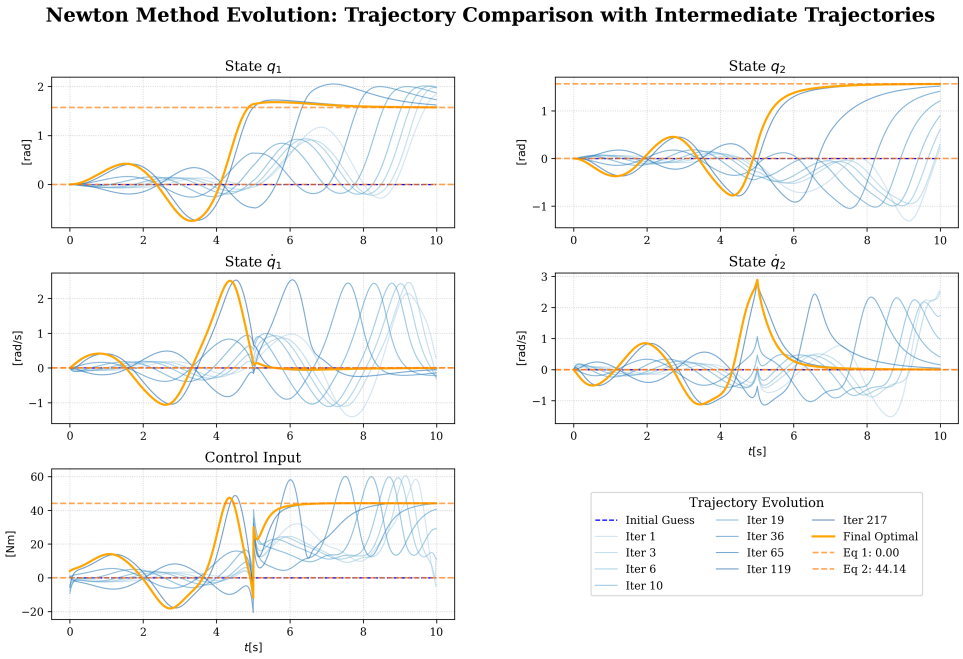


Figure 14: Newton's method applied to the Elbow Up maneuver using Stepwise reference. Note the control discontinuity manifesting as an infeasible torque spike.

3.2.2 Polynomial Interpolation and Dynamic Feasibility

Conversely, the **Polynomial (Poly 5)** reference offers a C^2-continuous kinematic guide, ensuring that position and velocity boundary conditions are respected. While this mitigates the initial gradient shock, the solver must still reconcile this kinematic path with the under-actuated dynamics of the flexible link.

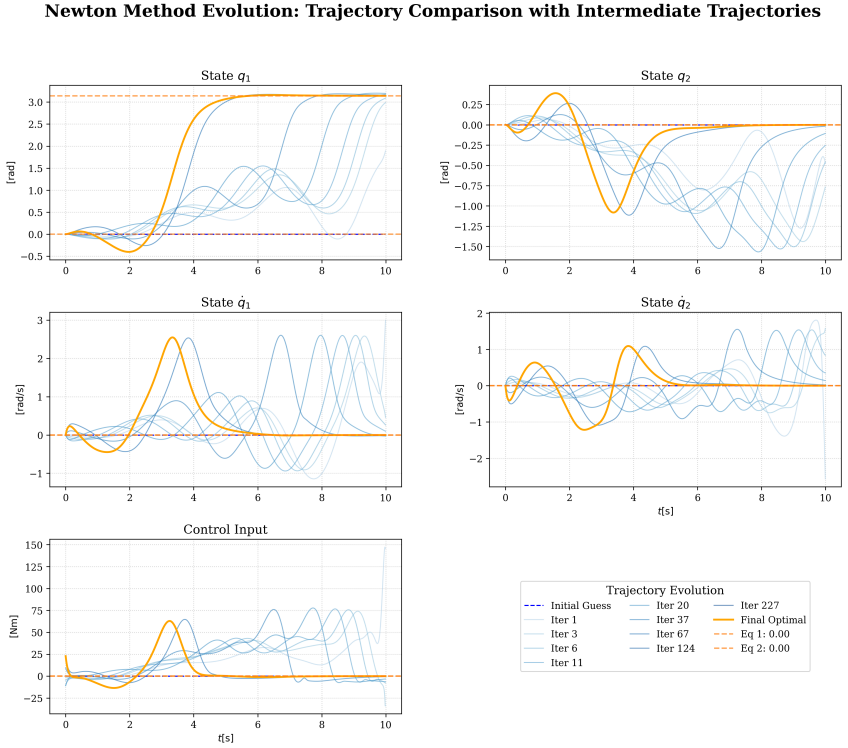


Figure 15: Optimization results: Poly 5 reference applied to the Swing Up task.

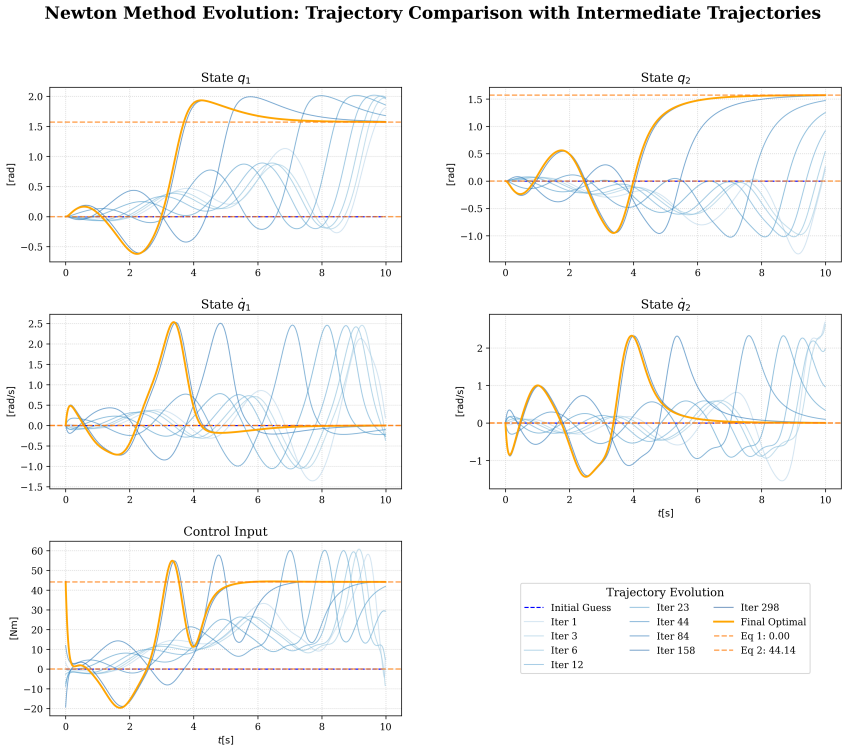


Figure 16: Optimization results: Poly 5 reference applied to the Elbow Up task.

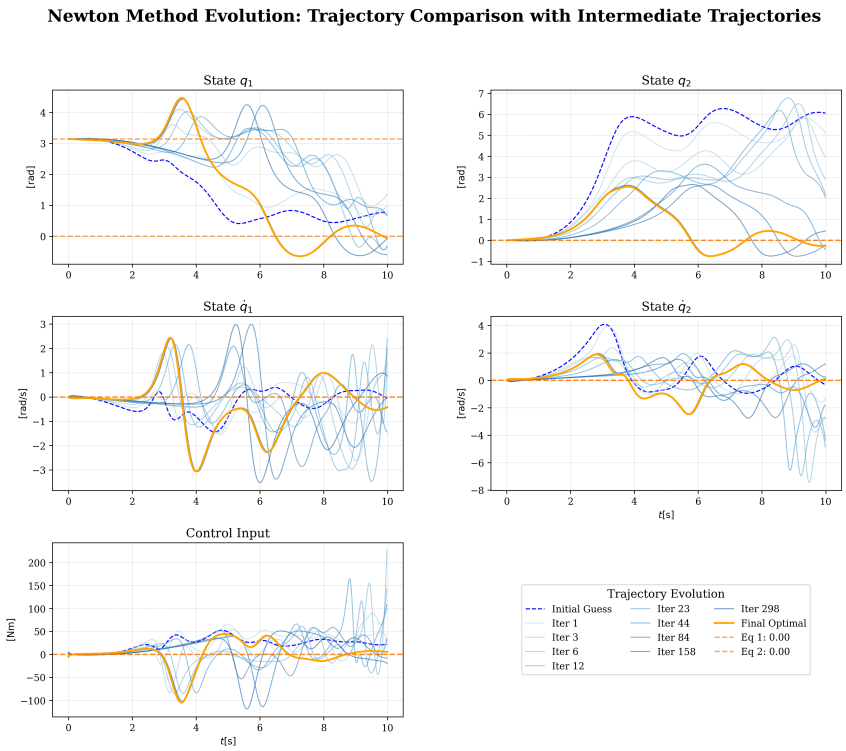


Figure 17: Optimization results: Poly 5 reference applied to the Swing Down task.

The results demonstrate that Newton's method successfully converges to accurate trajectories for upward maneuvers (Swing Up, Elbow Up). However, the algorithm struggles to generate valid trajectories for the Swing Down task, as evidenced by the deviation in Figure 17.

This dichotomy in performance is attributable to the fundamental difference in energy dy-

namics required for these maneuvers:

- **Energy Conversion (Upward Maneuvers):** In swing-up tasks, the control objective aligns with the natural conversion of kinetic energy into potential energy. As the links rise, gravity naturally decelerates the system. The polynomial reference effectively guides this energy transfer, requiring only corrective control efforts to maintain the path.
- **Energy Extraction (Downward Maneuvers):** In contrast, downward motions involve a release of potential energy, which accelerates the system. Due to the under-actuated nature of the second link and the low-friction environment, the system lacks a passive mechanism to dissipate this accumulated kinetic energy. The controller cannot brake the passive link directly; instead, it must actively extract energy from the system via dynamic coupling between the links. This requirement creates a complex, non-intuitive optimization landscape where simple polynomial initialization fails to provide a sufficient gradient towards the feasible solution.

3.3 Optimization Challenges

Applying the standard Newton method directly to complex, long-horizon maneuvers revealed several critical numerical and theoretical issues:

- **Non-Convex Optimization Landscape:** The trajectory manifold for complex maneuvers is highly non-linear, and the cost landscape is riddled with local minima. If the initial guess falls within the basin of attraction of a local minimum, representing a partial swing or a stalled state, the solver is unable to escape to the local optimum that guarantees the wanted trajectory.
- **Long Time Horizons and Sensitivity:** Extending the time horizon T exponentially increases the complexity of the solution space. Consequently, a long horizon increases the condition number of the Hessian matrix. Small changes in initial inputs propagate through the sensitivity matrix, causing massive deviations in terminal states and leading to numerical divergence.
- **Fixed Horizon Inflexibility:** A significant limitation of standard Newton's method is the rigidity of the fixed time horizon. If the chosen T is mismatched with the natural frequency of the maneuver, the optimization may be physically impossible to solve, leading to poor convergence.
- **Controllability and Terminal Cost:** Implementing a stabilizing terminal cost Q_T requires solving the Discrete Algebraic Riccati Equation (DARE). This necessitates the system linearization (A, B) at the target equilibrium to be stabilizable. In under-actuated configurations, kinematic singularities can cause rank deficiencies in the controllability matrix, rendering the DARE unsolvable.

  While standard controllability checks the rank of the Kalman controllability matrix $[B, AB, \ldots, A^{n-1}B]$, numerical conditioning is better assessed via the Popov-Belevitch-Hautus (PBH) test. A system is stabilizable if and only if:

$$\text{rank}([\lambda I - A \mid B]) = n \quad \forall \lambda \in \mathbb{C}, \quad |\lambda| \geq 1 \tag{9}$$

  In underactuated systems like in our case, kinematic singularities can cause this rank to drop (in our case we control only through coupling!). If the rank deficiency occurs at an unstable eigenvalue, the mode is *uncontrollable* and *unstable*. In such configurations, the Hessian becomes singular, and the Newton step is undefined.

- **Instability to Dwell Time:** We observed numerical instability when requiring the system to maintain an unstable equilibrium for a final time interval (dwell time). Consider the analogy of balancing a ball on a sharp peak: reaching the peak at time T requires precision, but remaining there requires perfect force cancellation over time. Any numerical error ϵ at $t < T$ is amplified by the unstable eigenvalues of the system evolution matrix (A^k). This creates a "narrow ridge" in the cost landscape where slight deviations result in massive penalties.
- **Linearization Validity:** The optimization relies on linearizing the non-linear dynamics around the current trajectory. If the update step is too large, the Taylor approximation breaks down, leading to valid descent directions becoming actual ascent directions in the true non-linear landscape.

3.3.1 Visual Analysis of the Local Minimum Trap

The issue of local minima is illustrated in Figure 18. Despite starting with a reasonable guess, the intermediate iterations cluster around a suboptimal path, and the optimization fails to reach the final equilibrium.

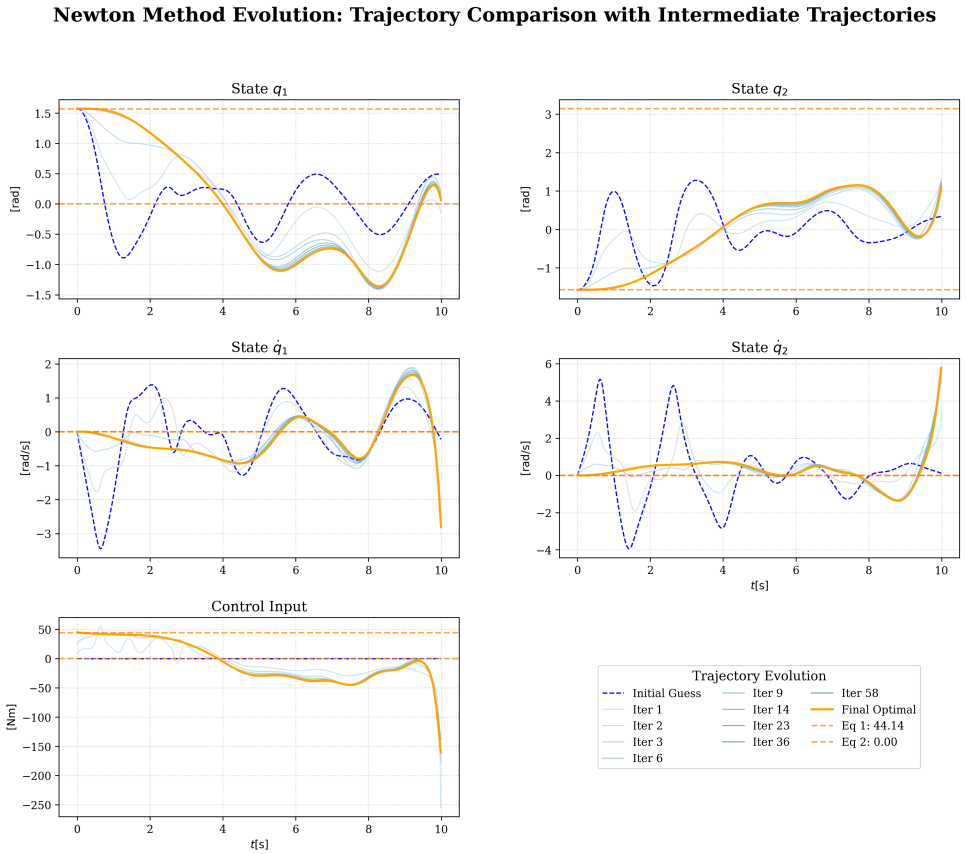


Figure 18: Illustration of the "Local Minimum Trap" where the solver converges to a suboptimal trajectory rather than the target equilibrium.

Crucially, standard convergence metrics can be misleading in this scenario. Figures 19 and 20 show that the cost function plateaus to a finite value and the descent direction norm approaches zero. This indicates that the solver has successfully found a mathematical minimum (gradient is zero), but it is a local minimum that fails to satisfy the physical task requirements. Nominal convergence does not imply task success.

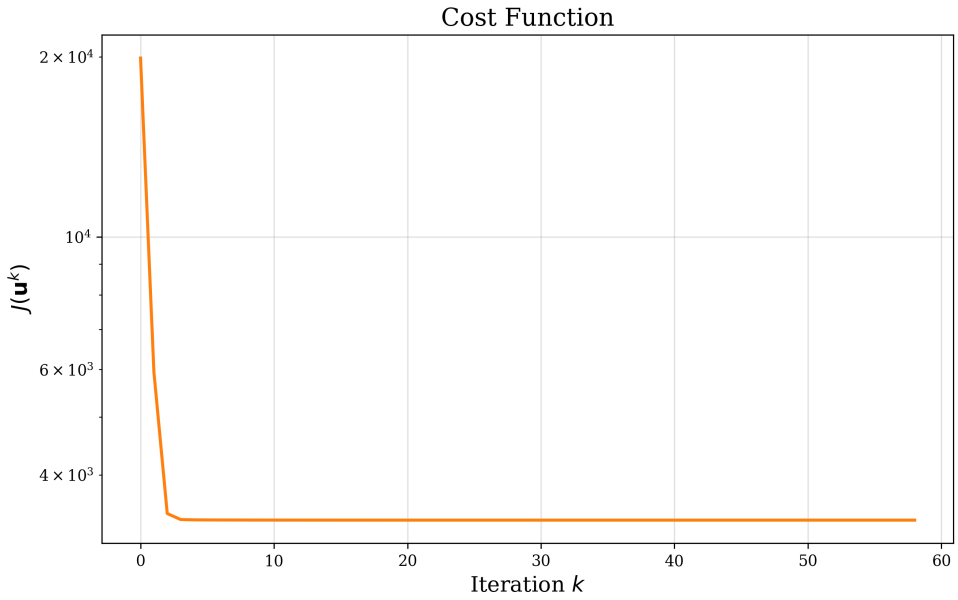


Figure 19: Cost function evolution during failure. The cost converges to a steady value, concealing the failure to reach the target.

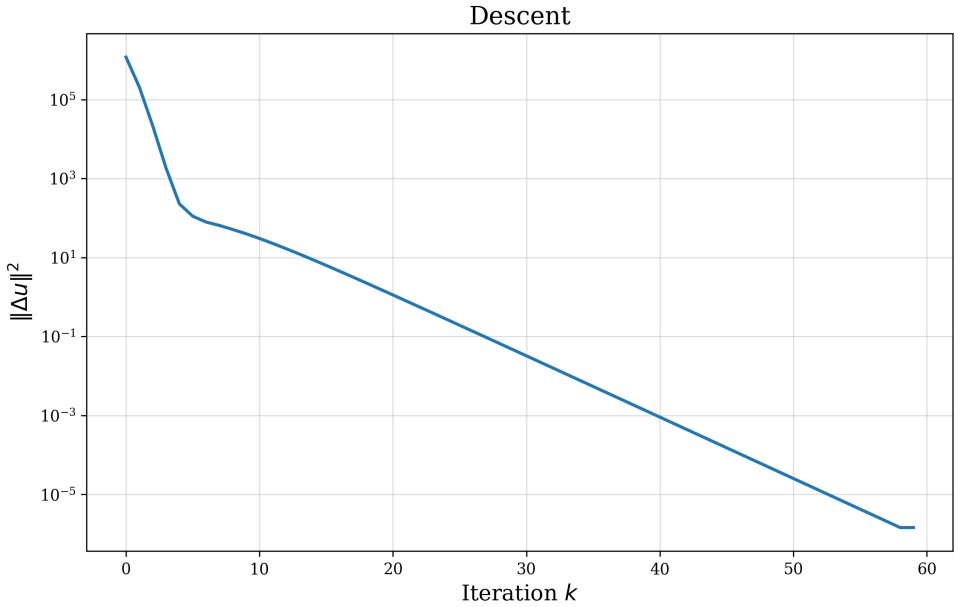


Figure 20: Descent direction norm. The algorithm mathematically converges (norm $\to 0$), but to an incorrect solution.

3.4 Proposed Solution: Dynamic Horizon Extension

To address the ill-conditioning and local minima problems described previously, we implemented a **Dynamic Horizon Extension** strategy. Instead of attempting to solve the optimization problem for the full duration T_{final} immediately, we solve a sequence of sub-problems with incrementally increasing horizon lengths. This strategy effectively conditions the optimization by ensuring the solver always operates within a valid local approximation of the system dynamics.

3.4.1 Theoretical Justification

1. **Conditioning:** For a short horizon T, the norm of the sensitivity matrix ($\prod A_t$) remains bounded. This results in a well-conditioned Hessian and a significantly larger basin of attraction for Newton's method.

2. **Warm Starting:** The optimal solution derived for horizon T serves as the initial guess for the extended horizon $T + \Delta T$. This "anchors" the subsequent optimization in a feasible region of the state space, maintaining proximity to physical constraints.

3. **Iterative Refinement:** As the horizon expands, the solver's task is reduced to refining the control inputs locally rather than discovering a global solution from scratch. This prevents the explosive divergence typically associated with global searches in ill-conditioned landscapes.

3.4.2 Algorithm Implementation

The iterative procedure is defined as follows:

1. **Initialize:** Set a short initial horizon $T_{curr} \ll T_{final}$ such that the DARE-based terminal cost effectively approximates the remaining cost-to-go.

2. **Converge:** Execute the Newton solver until the optimality criteria (gradient norm close to zero) are met for the current T_{curr}.

3. **Extend:** Use the computed trajectory $x^*(t)$ as the initial guess for the next phase. The horizon is extended by ΔT, and the reference tail is padded with the target equilibrium. The initial guess for the new segment is generated by evolving the system dynamics forward from the previous terminal state.

4. **Repeat:** Increment $T_{curr} \leftarrow T_{curr} + \Delta T$ and repeat the process until $T_{curr} \geq T_{final}$ and the terminal error satisfies $\|\mathbf{x}_T - \mathbf{x}_{target}\| < \epsilon$.

3.4.3 Numerical Analysis and Validation

The efficacy of the Dynamic Horizon Extension is visually demonstrated in the evolution of the system trajectories and optimization metrics.

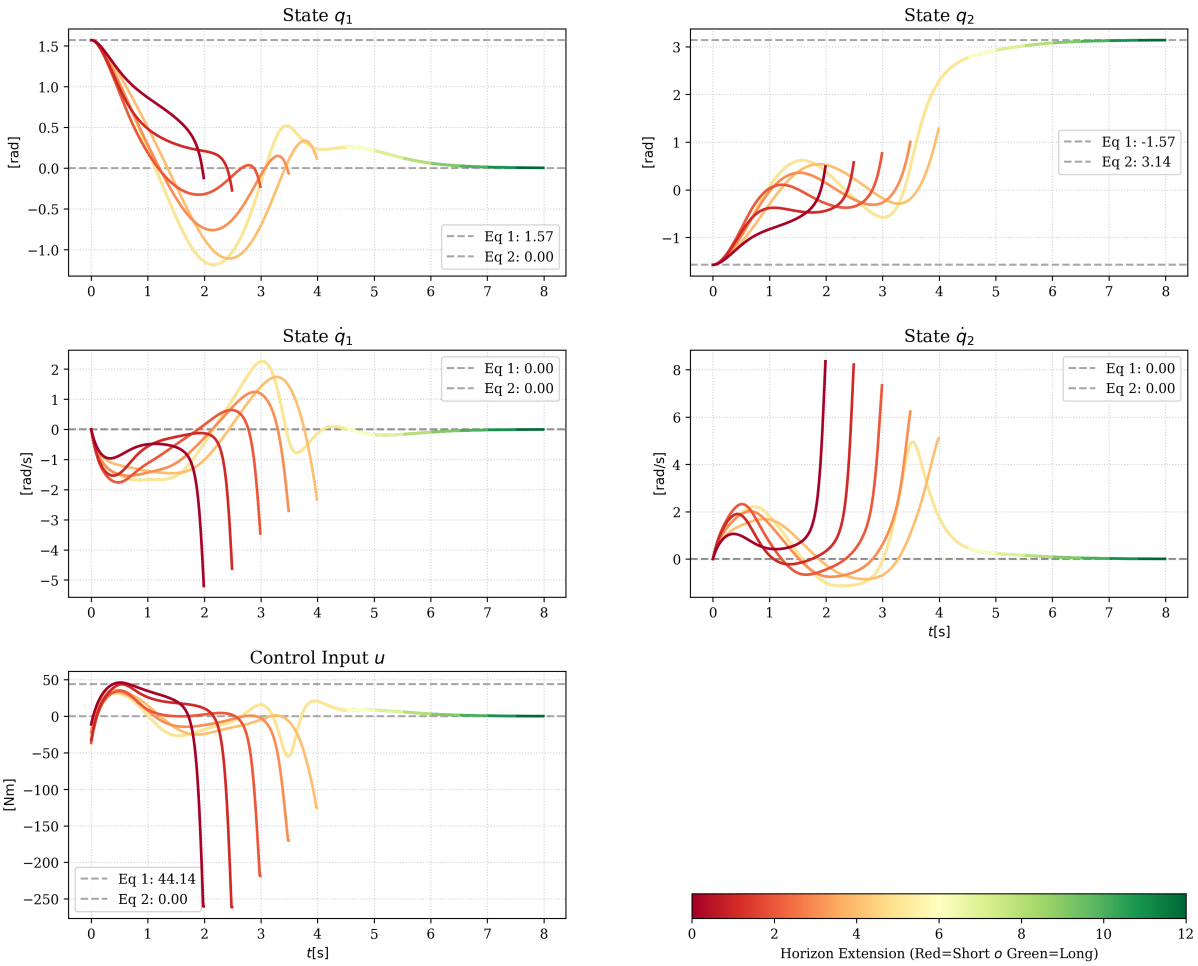


Figure 21: Evolution of the trajectory during the Dynamic Horizon Extension process. The system incrementally "learns" the complex maneuver.

Figure 21 illustrates the iterative learning process for a complex maneuver. Initially, the optimization explores suboptimal paths restricted by the short time horizon. As the horizon extends, the solver progressively uncovers the feasible manifold, guiding the trajectory into a "narrow channel" of state space that leads precisely to the unstable equilibrium. The abrupt changes in trajectory shape between phases indicate the solver escaping local basins of attraction that would otherwise trap a static-horizon optimization.

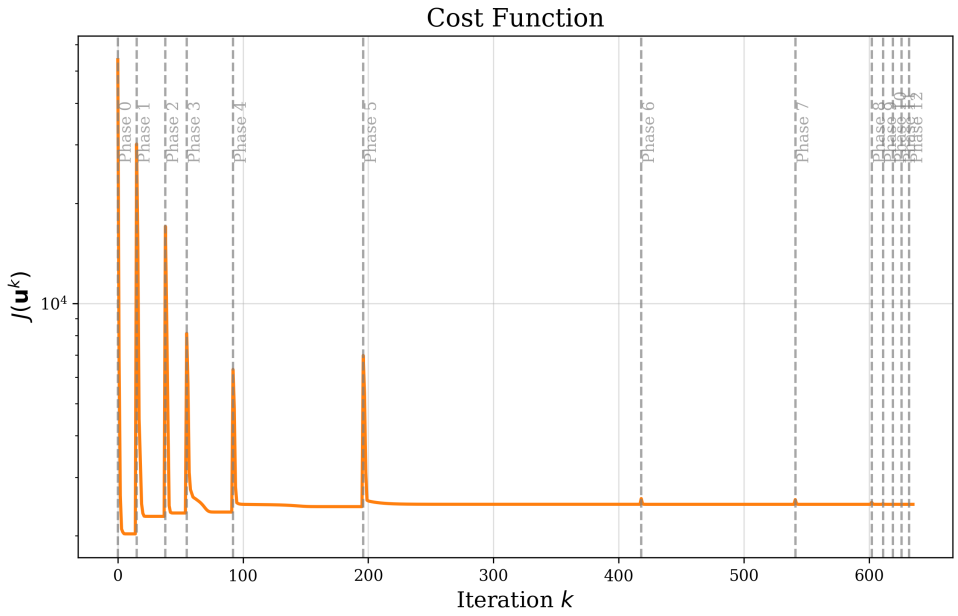


Figure 22: Step-wise evolution of the cost function. Each "step" corresponds to a horizon extension.

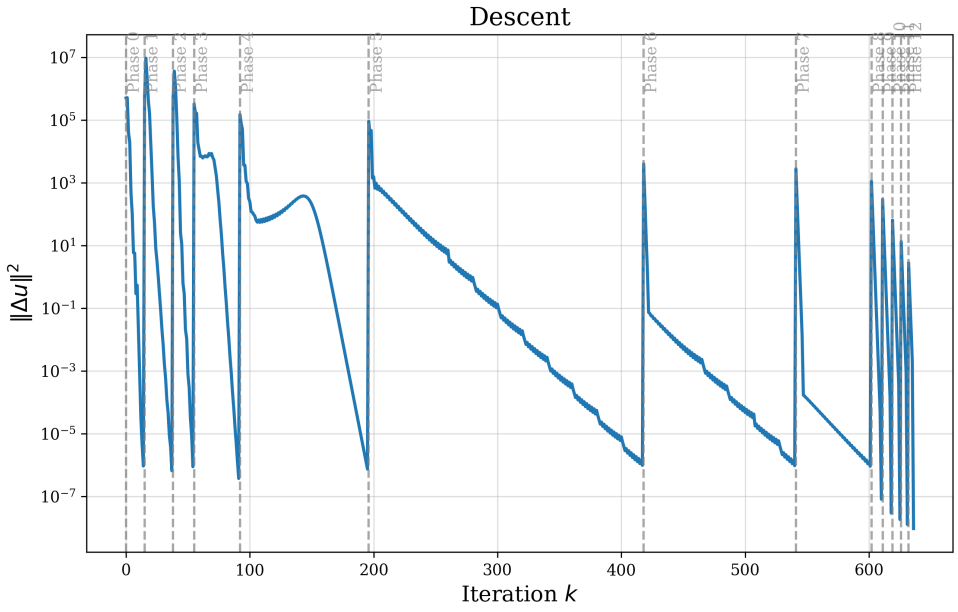


Figure 23: Norm of the descent direction. Spikes indicate the start of a new extension phase, followed by rapid convergence.

This behavior is further corroborated by the cost function evolution shown in Figure 22. It is observed that earlier, shorter solutions yield a lower total cumulative cost than the final "correct" solution. This result highlights the regularizing effect of the algorithm: a direct full-horizon optimization would likely converge to a lower-energy, locally optimal solution (like a partial swing) that fails to reach the target. By incrementally extending the horizon, we force the system to accept a higher-cost trajectory that satisfies the terminal boundary conditions.

The descent direction norm (Figure 23) confirms the stability of this approach. Each horizon extension introduces a perturbation, visible as a spike in the descent magnitude. However, because the system is warm-started from a feasible prior solution, the solver rapidly recon-

verges (descent norm $\to 0$) within a few iterations. This proves that the algorithm successfully transforms a difficult global search into a series of manageable local refinements.

The efficacy of the Dynamic Horizon Extension algorithm is further evidenced by the trajectories generated for the standard task set, presented in Figures 24 through **??**.

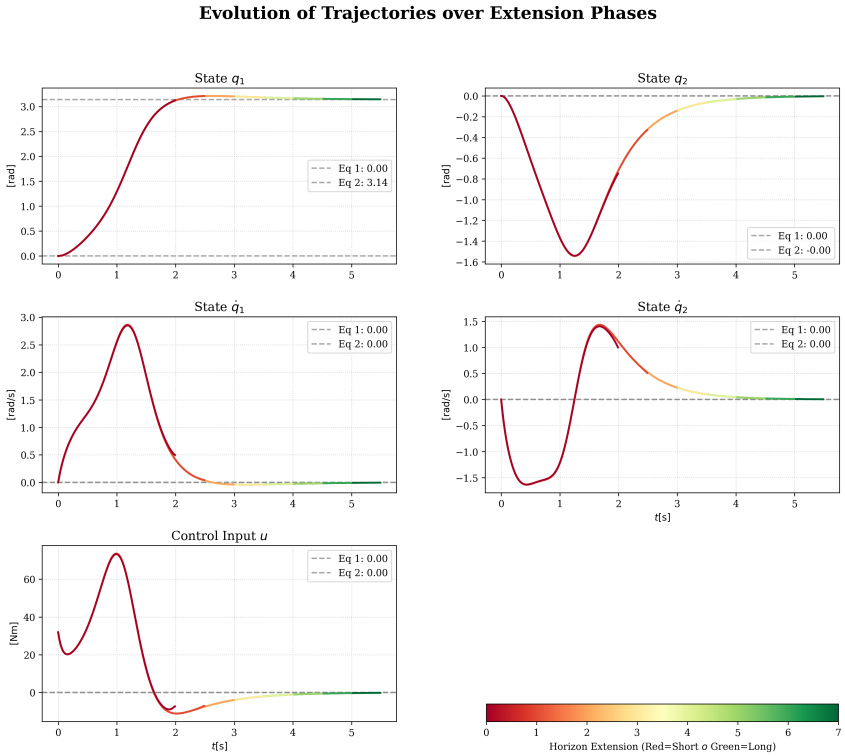


Figure 24: Optimization results: Poly 5 reference applied to the Swing Up task.

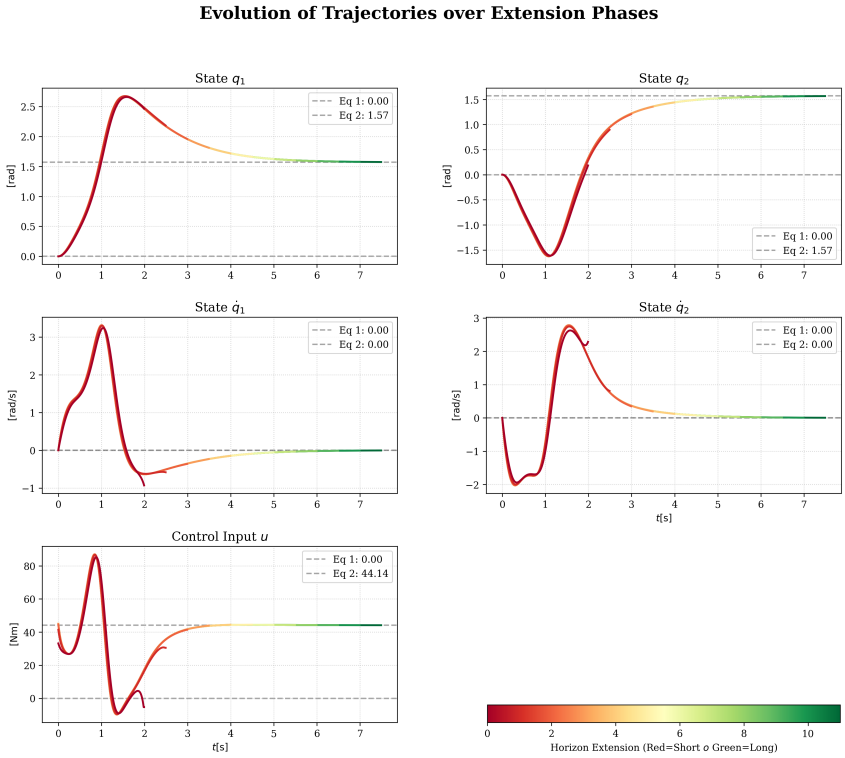


Figure 25: Optimization results: Poly 5 reference applied to the Elbow Up task.

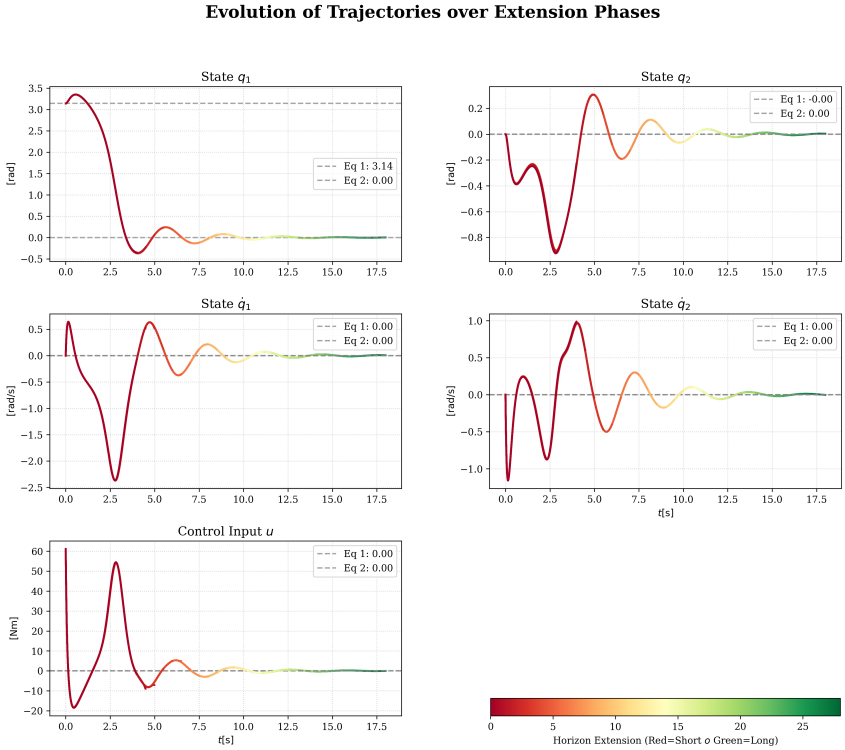


Figure 26: Optimization results: Poly 5 reference applied to the Swing Down task.

3.4.4 Convergence Analysis

The results highlight a clear distinction in the complexity of the solution manifolds for upward versus downward motions:

- **Upward Maneuvers (Swing Up, Elbow Up):** As observed in Figures 24 and **??**, the optimizer converges to a feasible trajectory almost immediately within the initial iterations. This rapid convergence suggests that the trajectory manifold for these energy-injection tasks is well-behaved and exhibits a quasi-convex structure in the vicinity of the polynomial reference. The solver easily locates a path that flows naturally toward the target equilibrium without requiring extensive temporal adjustments.
- **Downward Maneuver (Swing Down):** In contrast, the Swing Down task (Figure **??**) necessitates a substantial adaptation of the optimization problem. The algorithm autonomously extends the time horizon significantly, settling on a final duration approaching $T \approx 20$s. This behavior underscores the difficulty of the energy extraction phase; the system requires a longer time window to pump energy out via dynamic coupling while respecting actuation limits.

Crucially, this result validates the necessity of the incremental approach. A standard Newton method initialized with a fixed 20s horizon would inevitably diverge due to the severe ill-conditioning of the Hessian matrix and the accumulation of sensitivity errors over such a long duration. The Dynamic Horizon Extension avoids this by solving a sequence of well-conditioned sub-problems, allowing it to traverse a solution path that is inaccessible to the static solver.

4 Task 3: Trajectory Tracking via LQR

Once an optimal trajectory (x^*, u^*) is generated, we designed a **Linear Quadratic Regulator** (LQR) to stabilize the system around this trajectory in the presence of **disturbances**.
The main objective is to **minimize deviations** from the reference while ensuring **smooth control actions**, also ensuring stability at the end of the maneuver is **maintained or recovered**.

4.1 Linear Time-Varying (LTV) LQR

Since the reference trajectory is not a static point, we linearized the system dynamics along the trajectory, resulting in time-varying matrices A_t and B_t:

$$\delta x_{t+1} = A_t \delta x_t + B_t \delta u_t \tag{10}$$

The feedback law is computed as $u_t = u_t^* + K_t(x_t - x_t^*)$, where K_t is obtained by solving the Riccati equation backward in time. The input is therefore composed of the nominal **feedforward** term u_t^* and a **feedback** correction based on the state deviation.

4.2 Implementation Details

- **Perturbations:** To evaluate the controller's basin of **attraction**, we initialized the system with stochastic perturbations ($\epsilon \sim U[-0.05, 0.05]$) and then, for each run, we decided to apply the following initial disturbance:

  $$x_0 = x_0^* + \epsilon \cdot p_{\text{scale}} \cdot rr \tag{11}$$

  Where rr is the specific run, and p_{scale} is a scaling factor. By scaling these perturbations across multiple runs, we observed the LQR's ability to recover the nominal path even when the initial state deviated significantly from the optimal starting point.

- **Horizon Extension and Terminal Stability:** To ensure consistent behavior between the trajectory generation and tracking phases, the terminal cost matrix Q_T was defined as the **solution to the Discrete Algebraic Riccati Equation** (DARE) computed at the final equilibrium point.
  This choice effectively incorporates an infinite-horizon cost-to-go into the finite-horizon LQR framework, allowing the controller to **transition from a trajectory tracker into a steady-state regulator**.
  As shown in the extended simulations, the LQR continues to operate beyond the nominal time T to demonstrate the *asymptotic stability* of the closed-loop system.
  It is important to note that the validity of this terminal regulation relies on the **final trajectory state being a true equilibrium point**.
  This ensures that the linearized dynamics used for the DARE remain a faithful approximation of the nonlinear system, allowing the LQR to stabilize the robot within the neighborhood of the target pose even in the presence of residual perturbations.

- **Cost Function:** The weighting matrices Q, R, and Q_T were chosen to **balance state deviation penalties and control effort**, tuned through simulations to achieve desired performance.
  In this case we used similar or lower R to give more flexibility to recover from disturbances but, given given the flexible nature of the robot, high-gain tracking ($Q \gg R$) could **excite high-frequency oscillations** or lead to saturation in a real-world scenario.

4.3 Simulation Results

The results (see Figure 27, 29, 31, 33) demonstrate that the LTV-LQR effectively rejects initial disturbances and tracks the complex optimal maneuver. It is possible to observe how the final stabilization of the second term is left onto the first arm, which oscillates to neglect the perturbations on the second link, and only later it will stabilize.

In this simulation 5 second of stand-still at the final equilibrium has been added for every task.

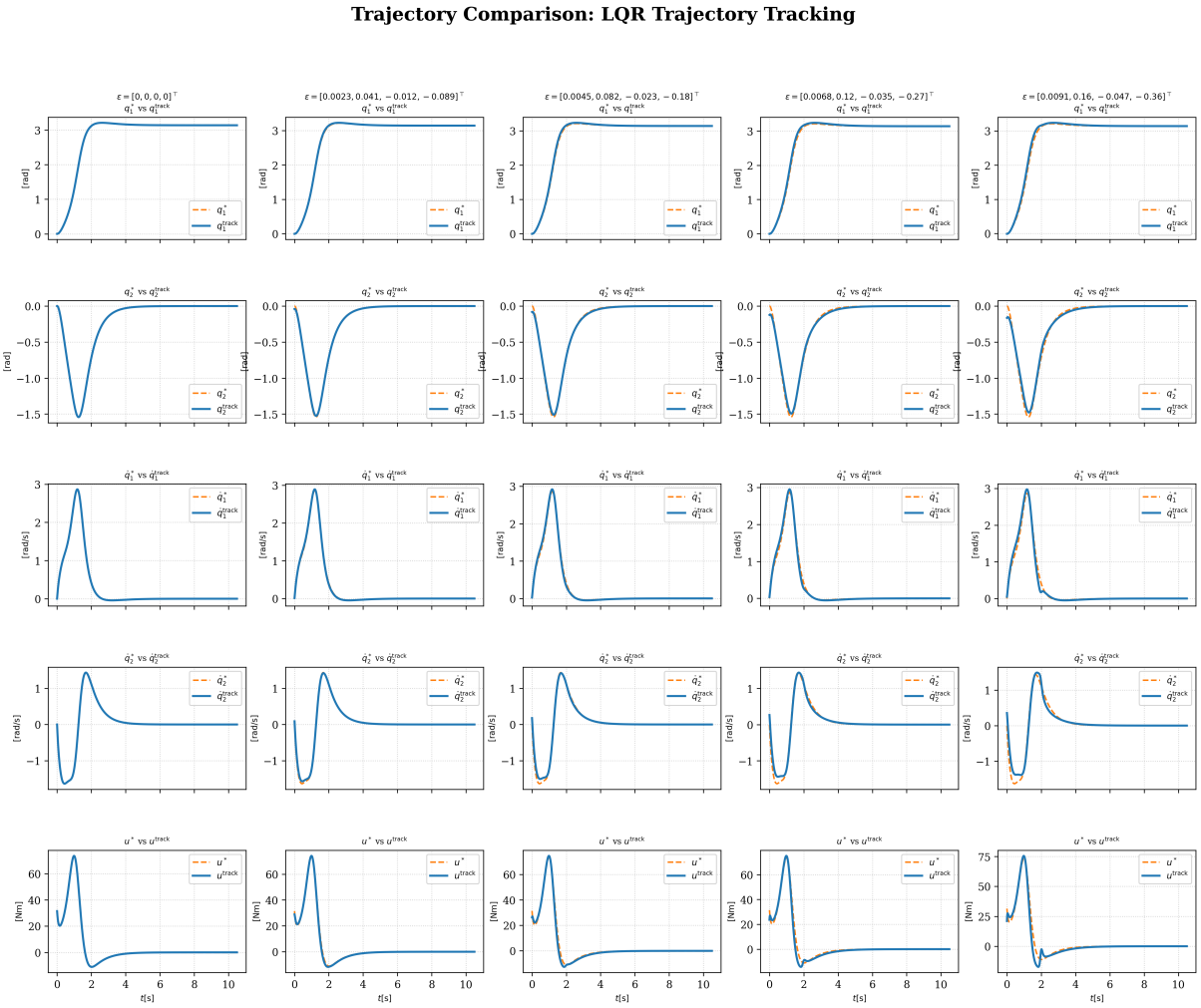


Figure 27: Trajectory tracking on a swing-up maneuver

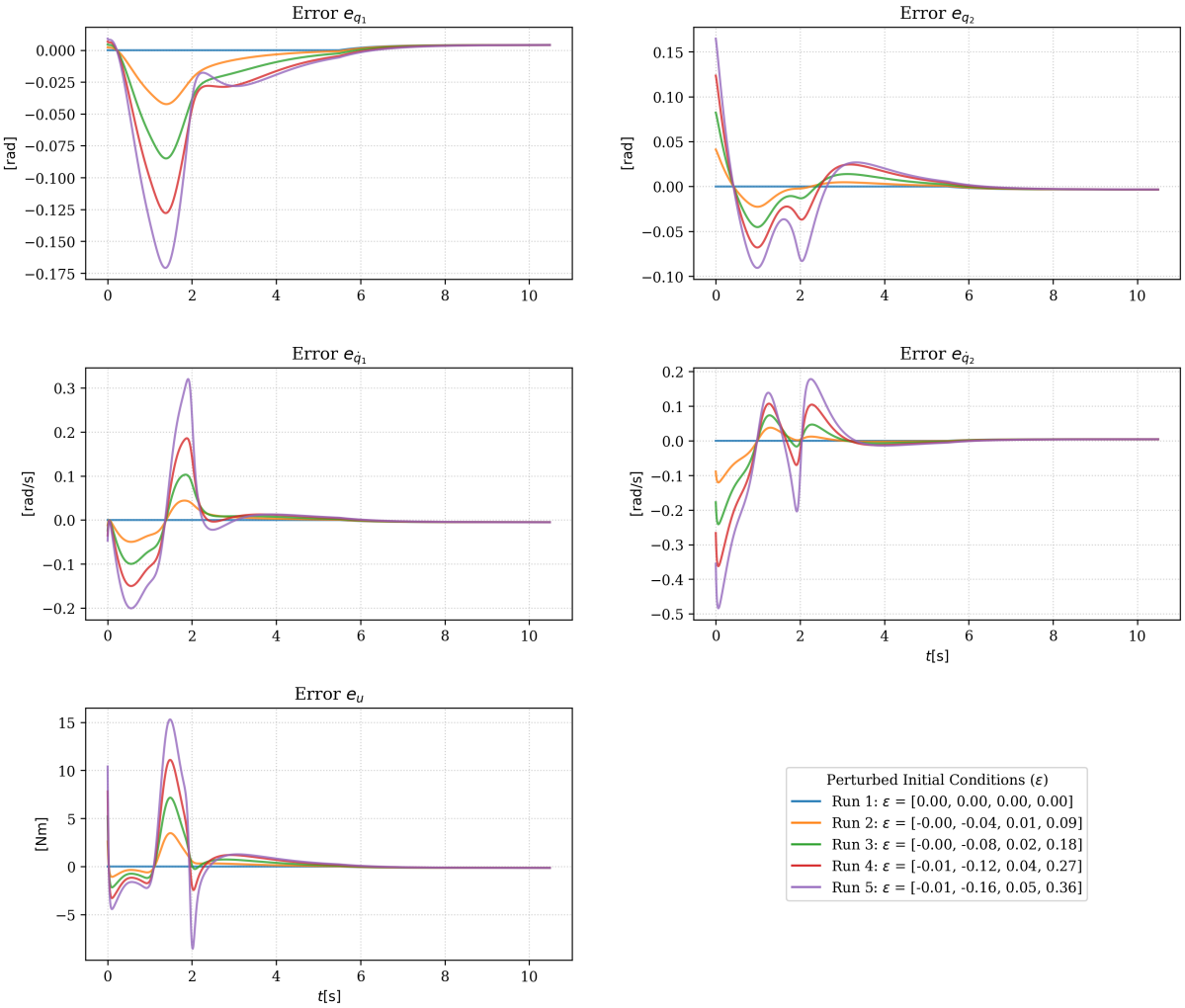


Figure 28: Error analysis during LQR tracking on a swing-up maneuver

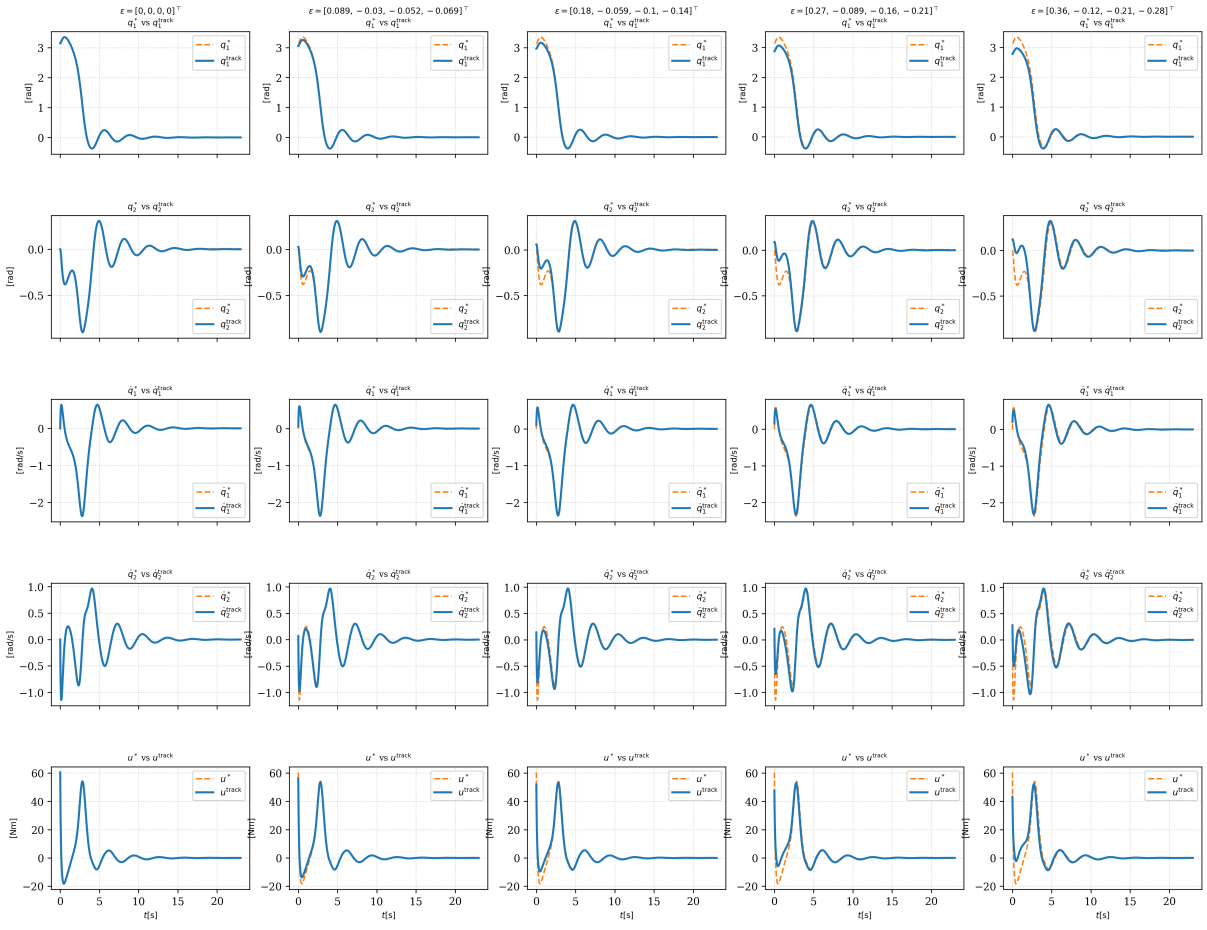


Figure 29: Trajectory tracking on a swing down maneuver

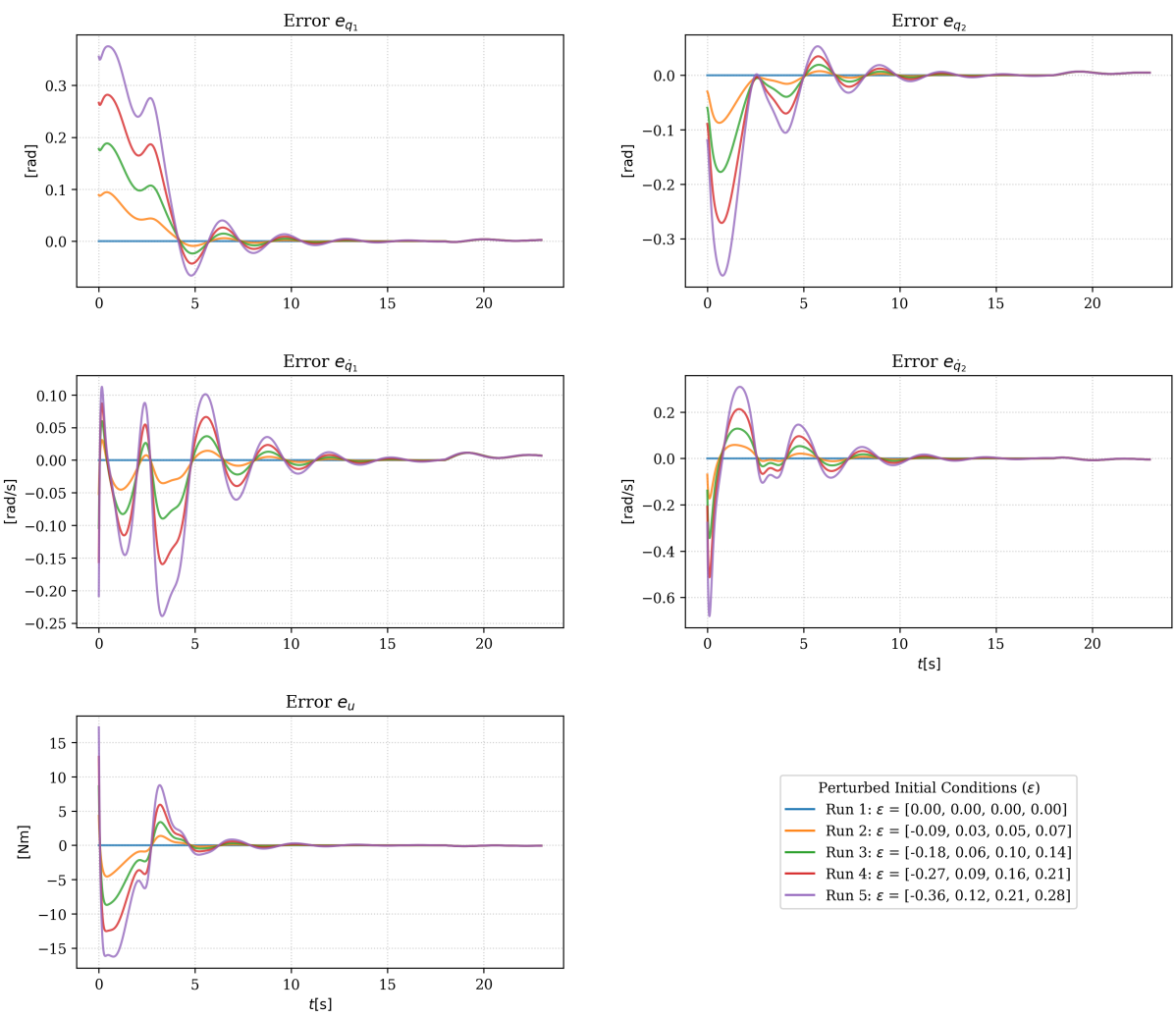


Figure 30: Error analysis during LQR tracking of a swing down maneuver

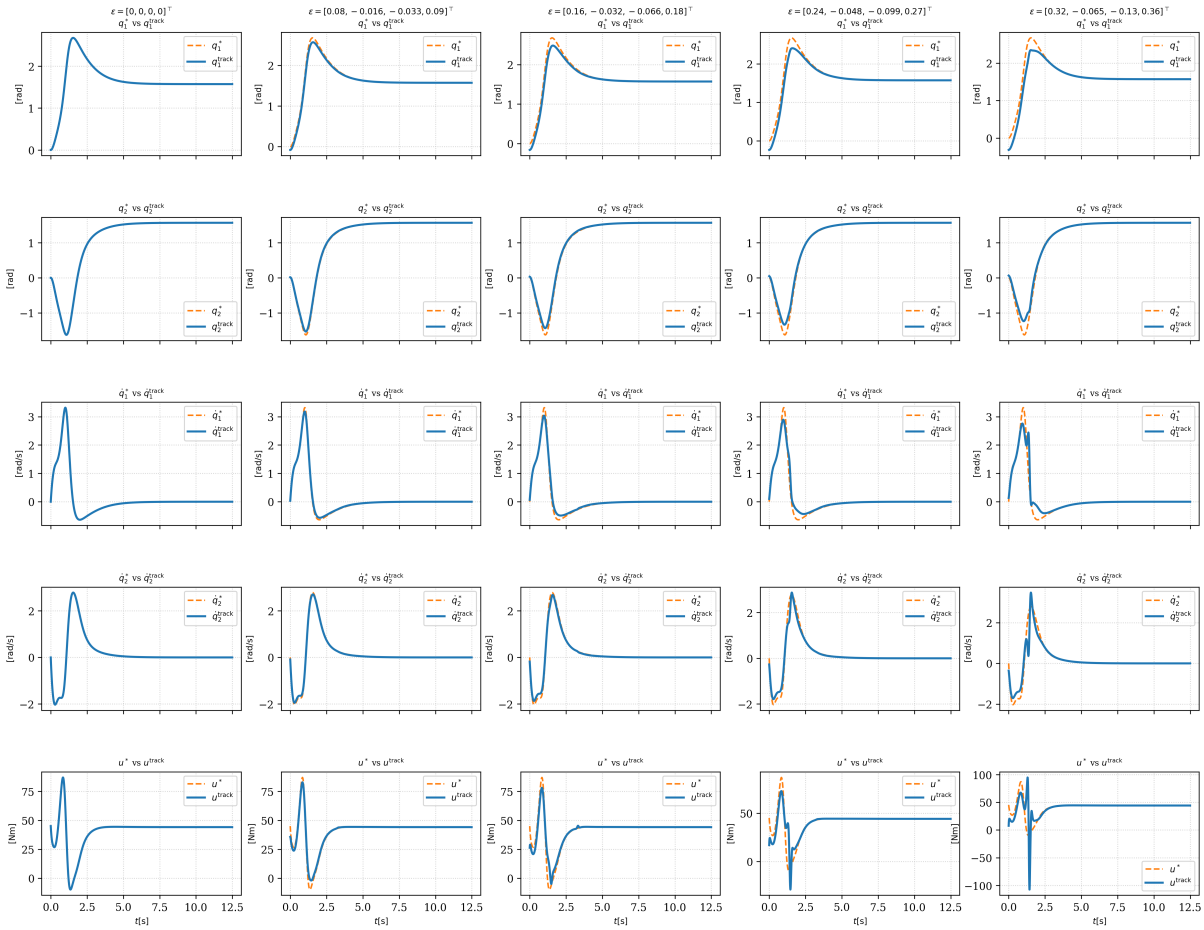


Figure 31: Trajectory tracking on a elbow up maneuver

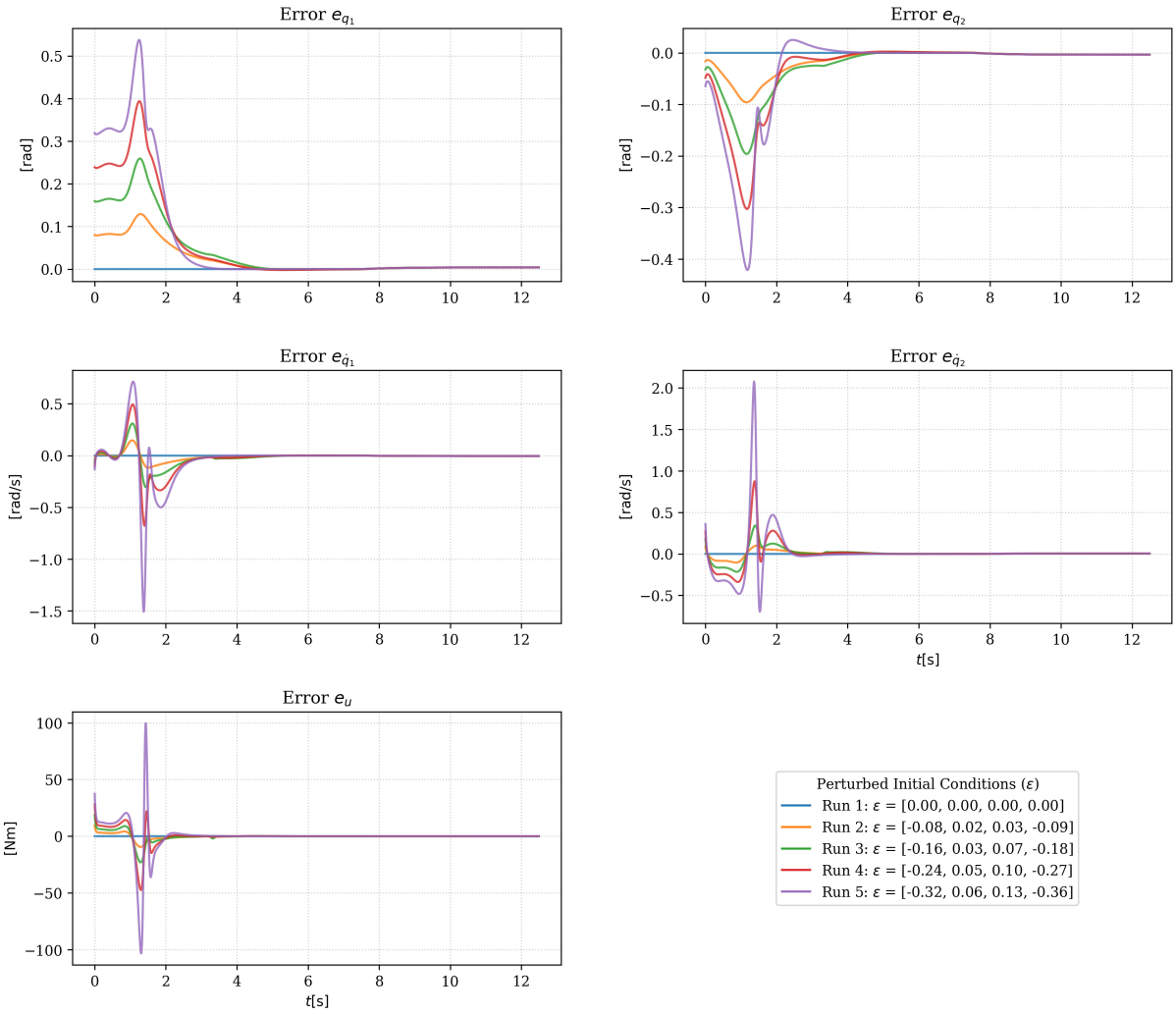


Figure 32: Error analysis during LQR tracking of elbow up maneuver

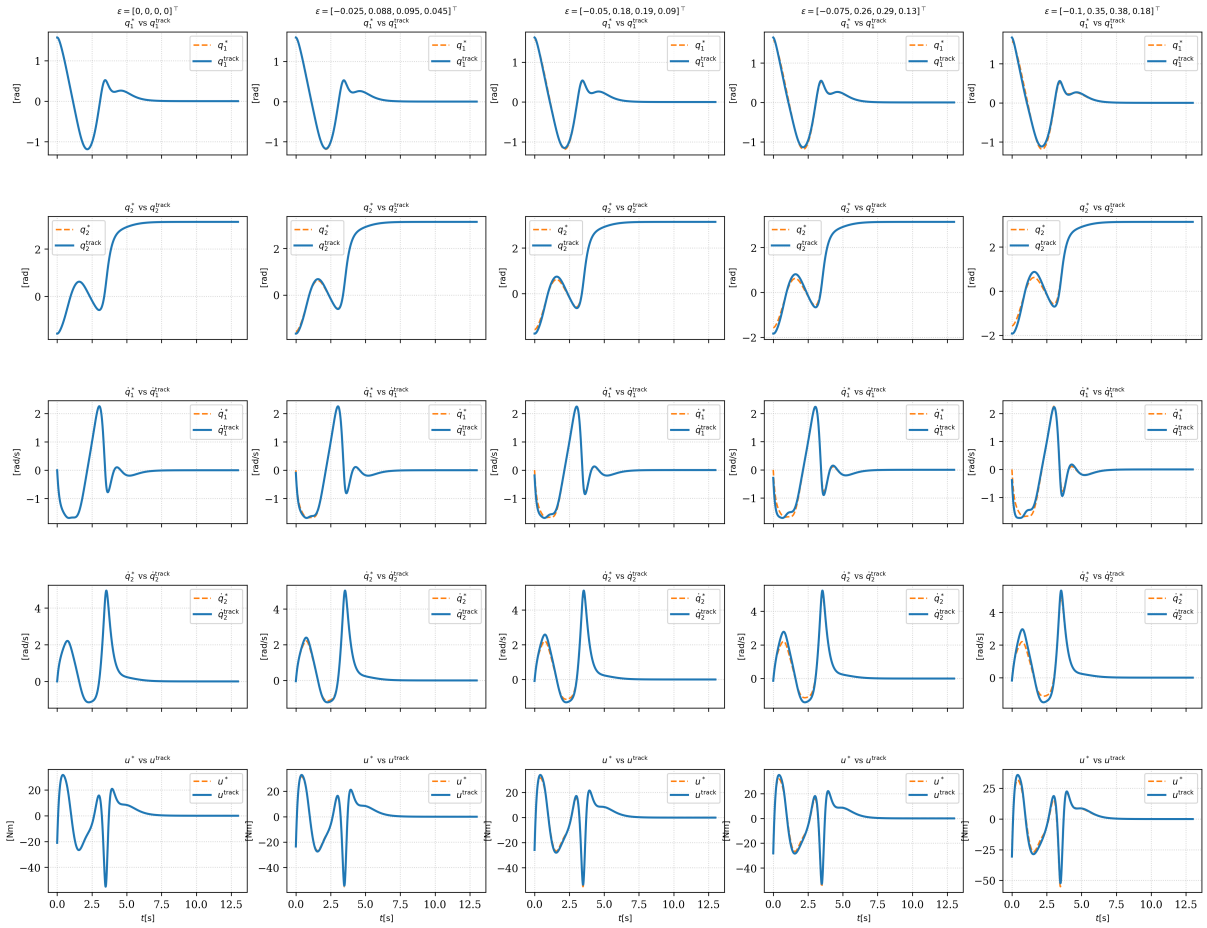


Figure 33: Trajectory tracking on a complex maneuver

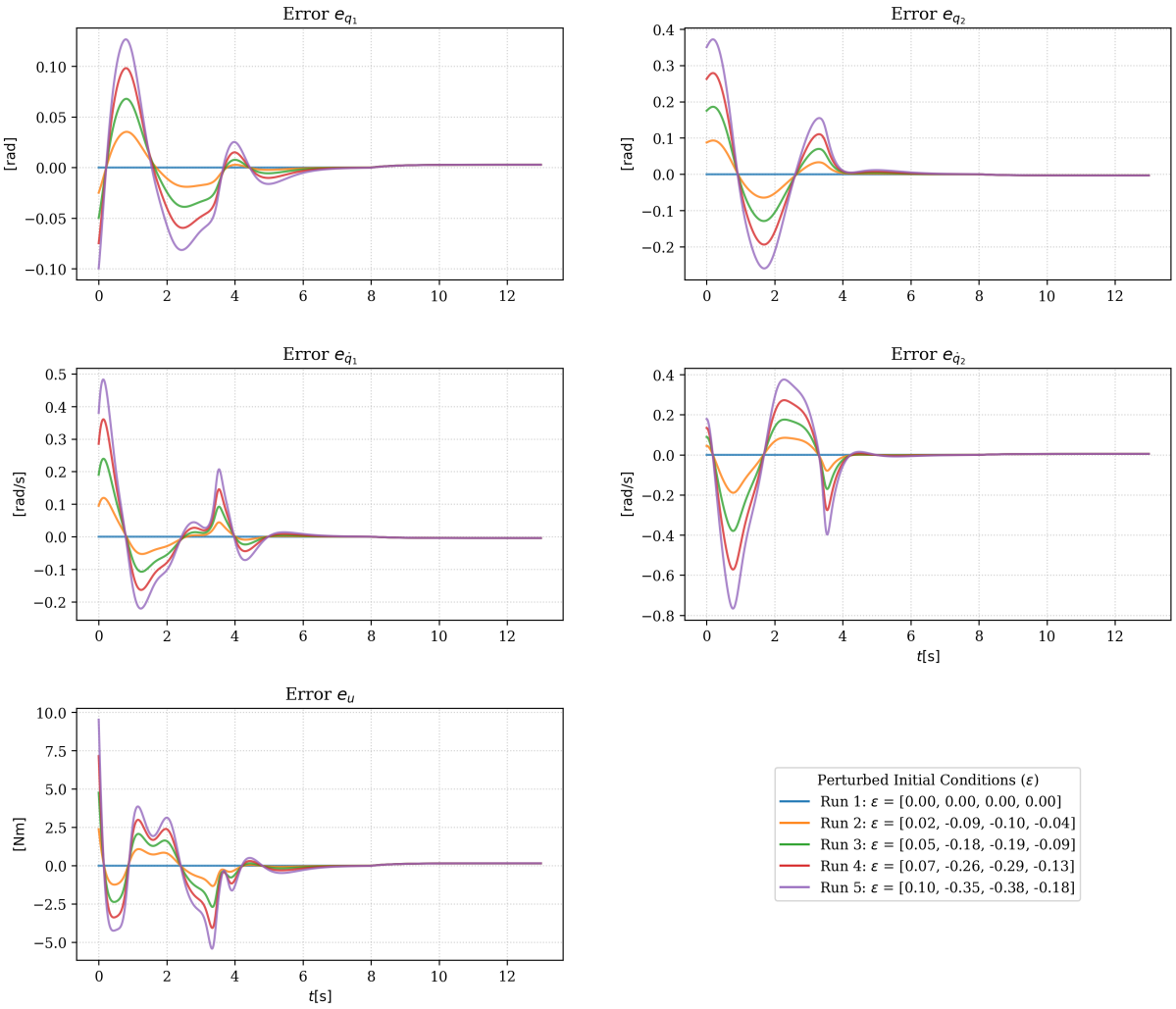


Figure 34: Error analysis during LQR tracking of complex maneuver

5 Task 4: Trajectory Tracking via MPC

While LQR is effective for unconstrained tracking, it **cannot explicitly handle actuator saturation**. We implemented a linear **Model Predictive Control** (MPC) scheme to address this.

5.1 MPC Formulation

At each time step t, we solve a finite-horizon optimization problem:

$$\begin{aligned} \min_{\Delta u} \quad & \sum_{t=\tau}^{\tau+T_{pred}} \left(\|\Delta x_\tau\|_Q^2 + \|\Delta u_\tau\|_R^2\right) + \|\Delta x_{T_{pred}}\|_{Q_T}^2 \\ \text{s.t.} \quad & \Delta x_{t+1} = A_t \Delta x_t + B_t \Delta u_t, \quad \forall t = \tau, ..., \tau + T_{pred} \\ & u_{min} \leq u_t^* + \Delta u_t \leq u_{max}, \quad \forall t = \tau, ..., \tau + T_{pred} \\ \Delta x_t = \Delta x_t^{meas} & \end{aligned}$$

where T_{pred} is the prediction horizon.

5.2 Key Features

- **Constraints:** We imposed hard constraints on the input torque ($u \in [u_{min}, u_{max}]$) to **respect physical motor limits**, which LQR ignores.
  In particular, we set $u_{max} = \max(|u^*|)$ and $u_{min} = -u_{max}$, so as **not to exceed the torque needed to complete the nominal trajectory** with no disturbances; managing to find a way through thanks to the prediction window and the optimization.
- **Receding Horizon:** Only the first control input is applied, and the optimization is repeated at the next step. Furthermore we opted for a prediction horizon T_{pred}
  that could serve as a **good compromise between performance and computational load**. In particular, we set $T_{pred} = \min(100, \frac{\text{len}(time)}{2})$.
- **Perturbations:** Similar to the LQR case, we introduced random initial state perturbations to evaluate robustness.
- **Solver:** We utilized CasADi with IPOPT to solve the constrained QP/NLP at each step.

5.3 Simulation Results

The MPC controller showed superior performance in scenarios where large perturbations would cause an LQR controller to request infeasible torques.
Moreover, the ability to handle constraints makes MPC a versatile choice.
Down below some results are shown (see Figure 35, 39, 37, 41). One can notice how, in some cases, the knowledge of the future helps in recovering the reference trajectory and final equilibrium, and other cases in which **the window and/or the constraints do not allow the controller to return to the nominal path** (as in Figure 35). Above all, the controller **never exceeds the maximum torque defined**, showing the effectiveness of the constraints in the optimization problem.

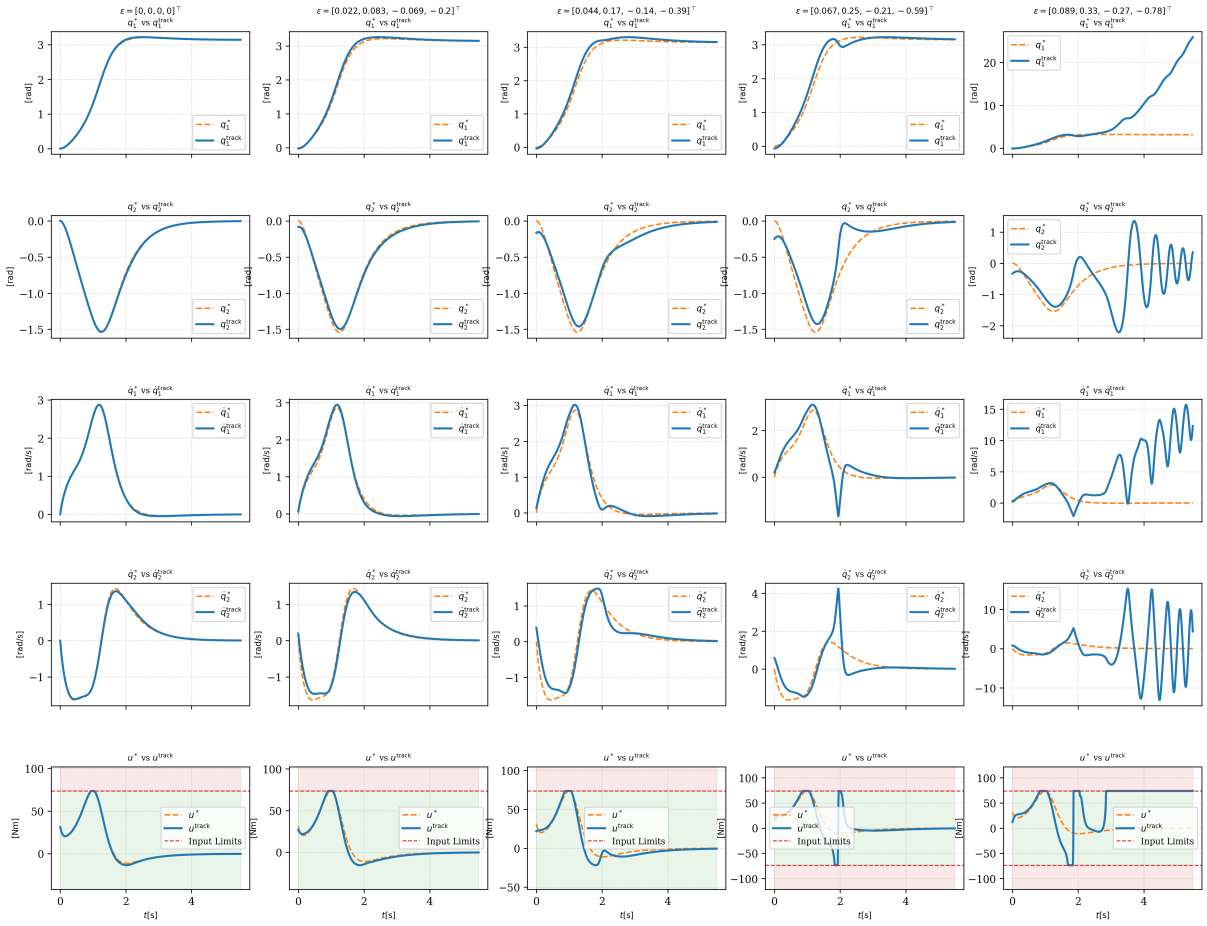


Figure 35: Trajectory tracking on a swing up maneuver

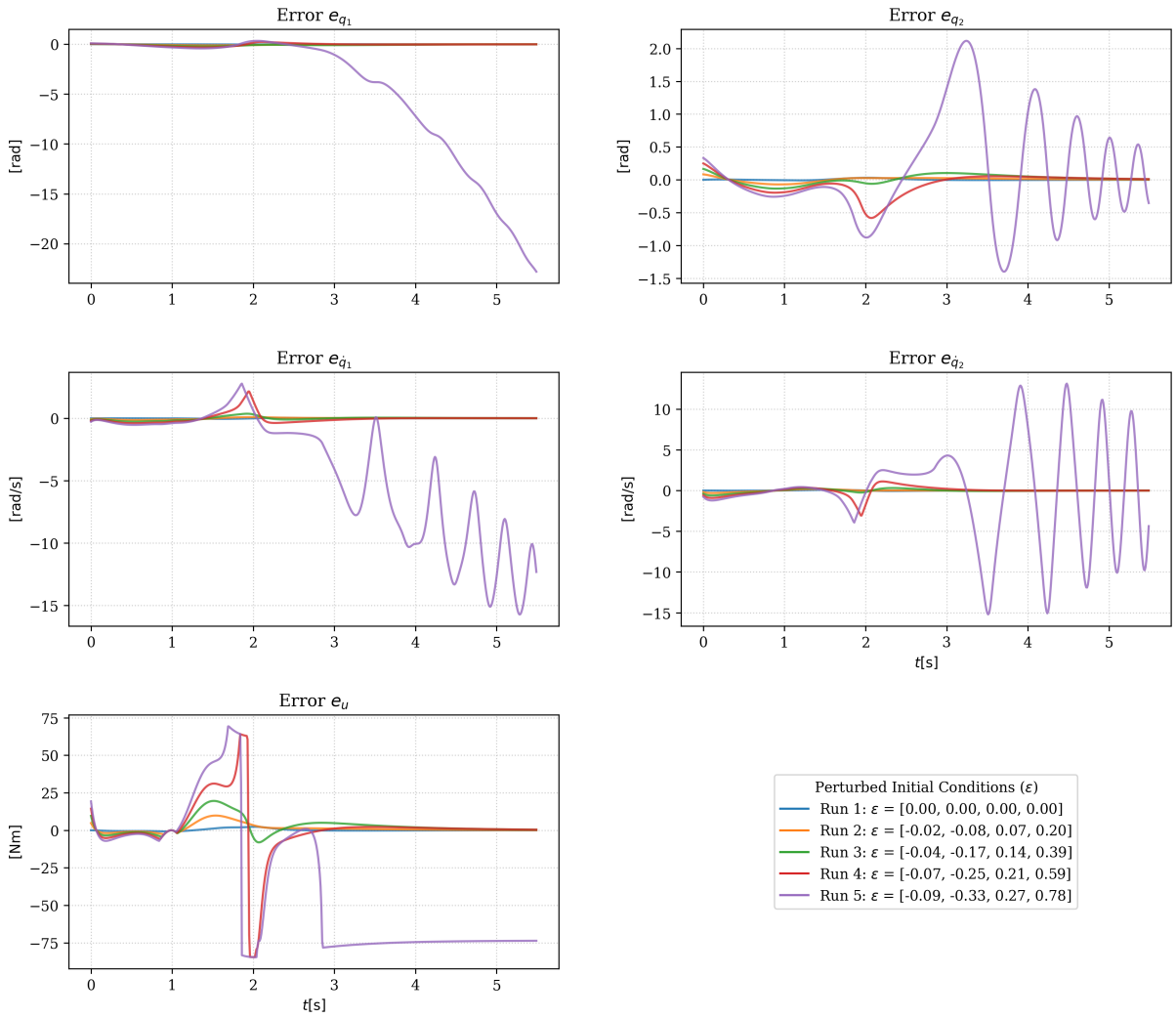


Figure 36: Error analysis during MPC tracking of swing up maneuver

Figure 37: Trajectory tracking on a swing down maneuver

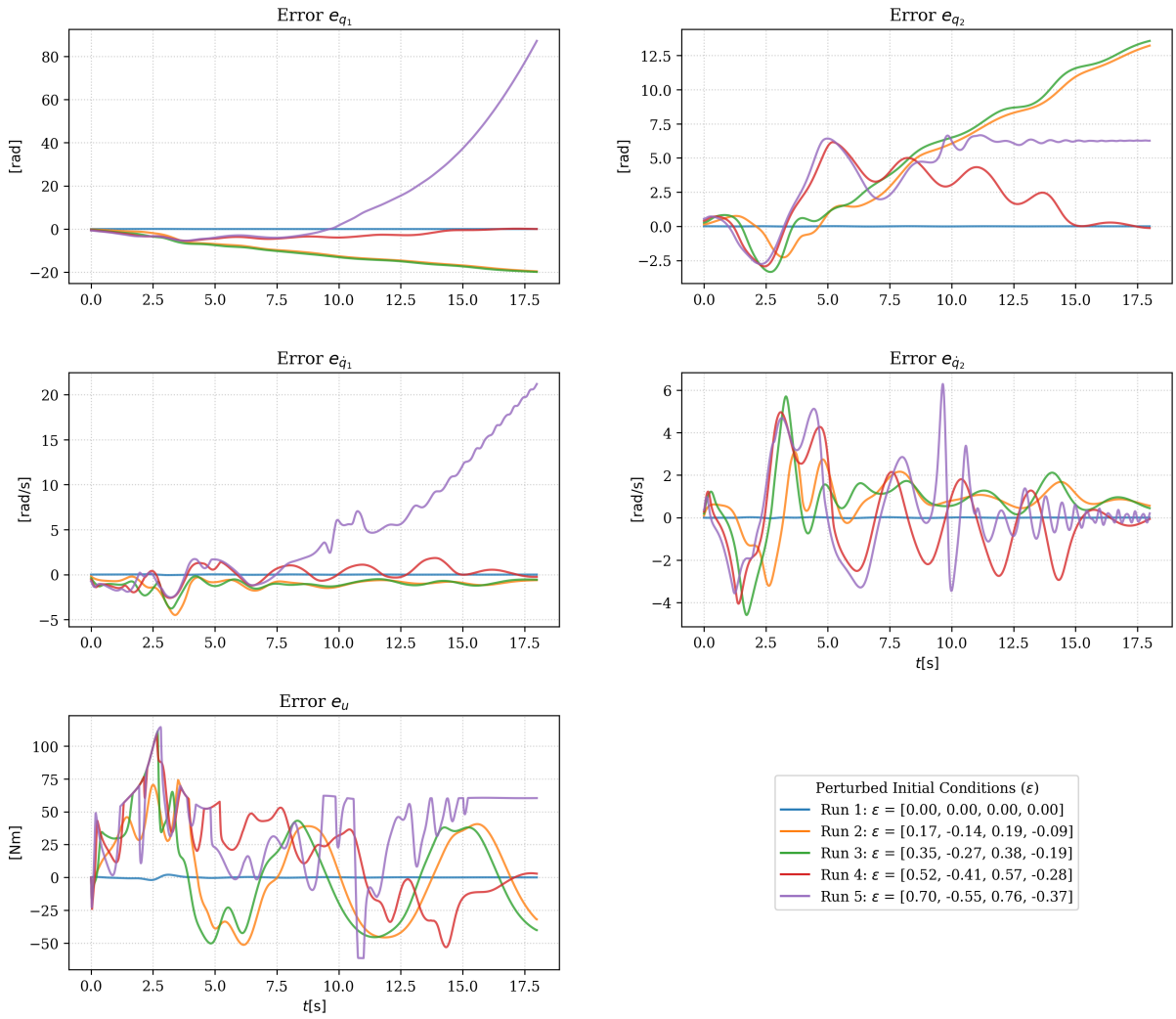


Figure 38: Error analysis during MPC tracking on a swing down maneuver

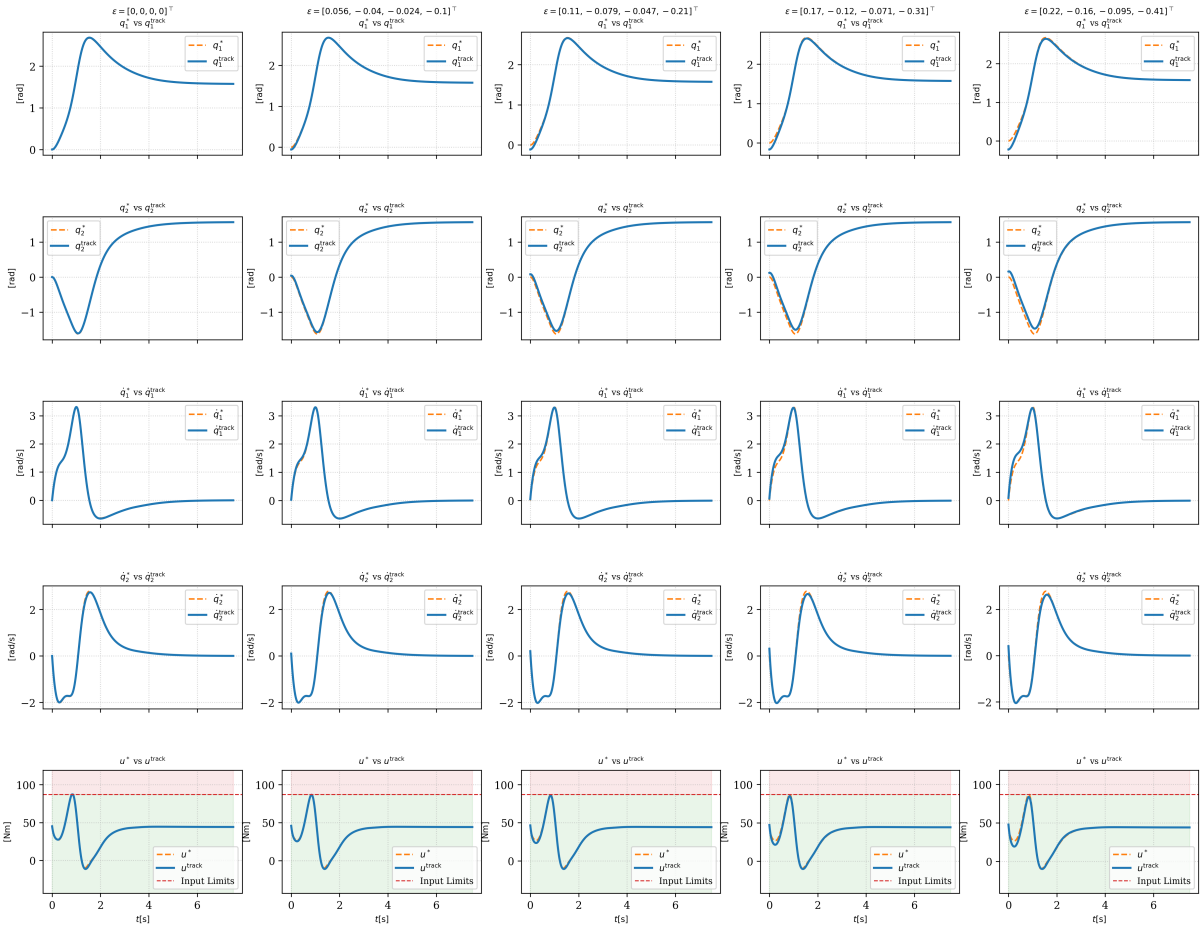


Figure 39: Trajectory tracking on a elbow up maneuver

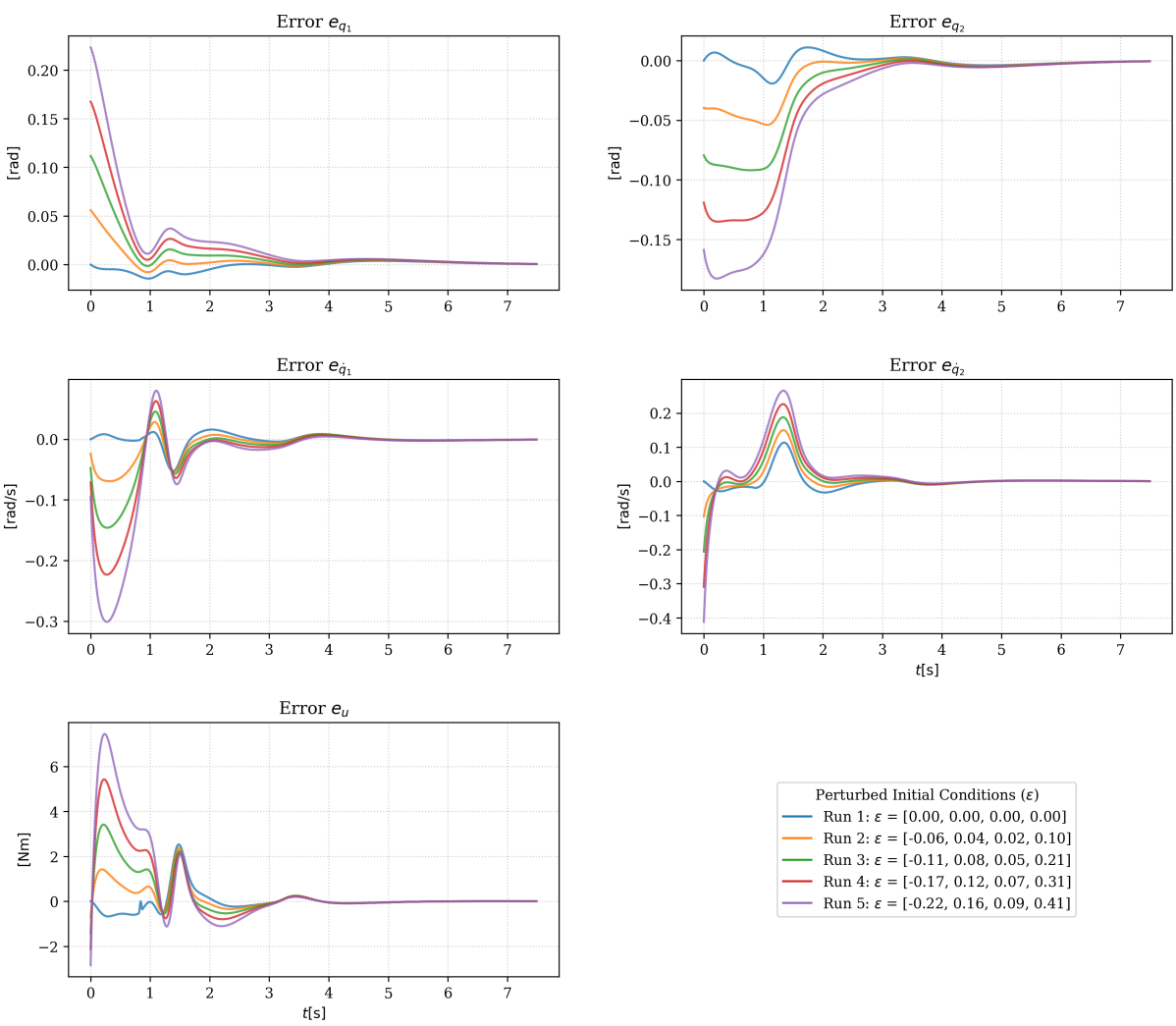


Figure 40: Error analysis during MPC tracking of a elbow up maneuver

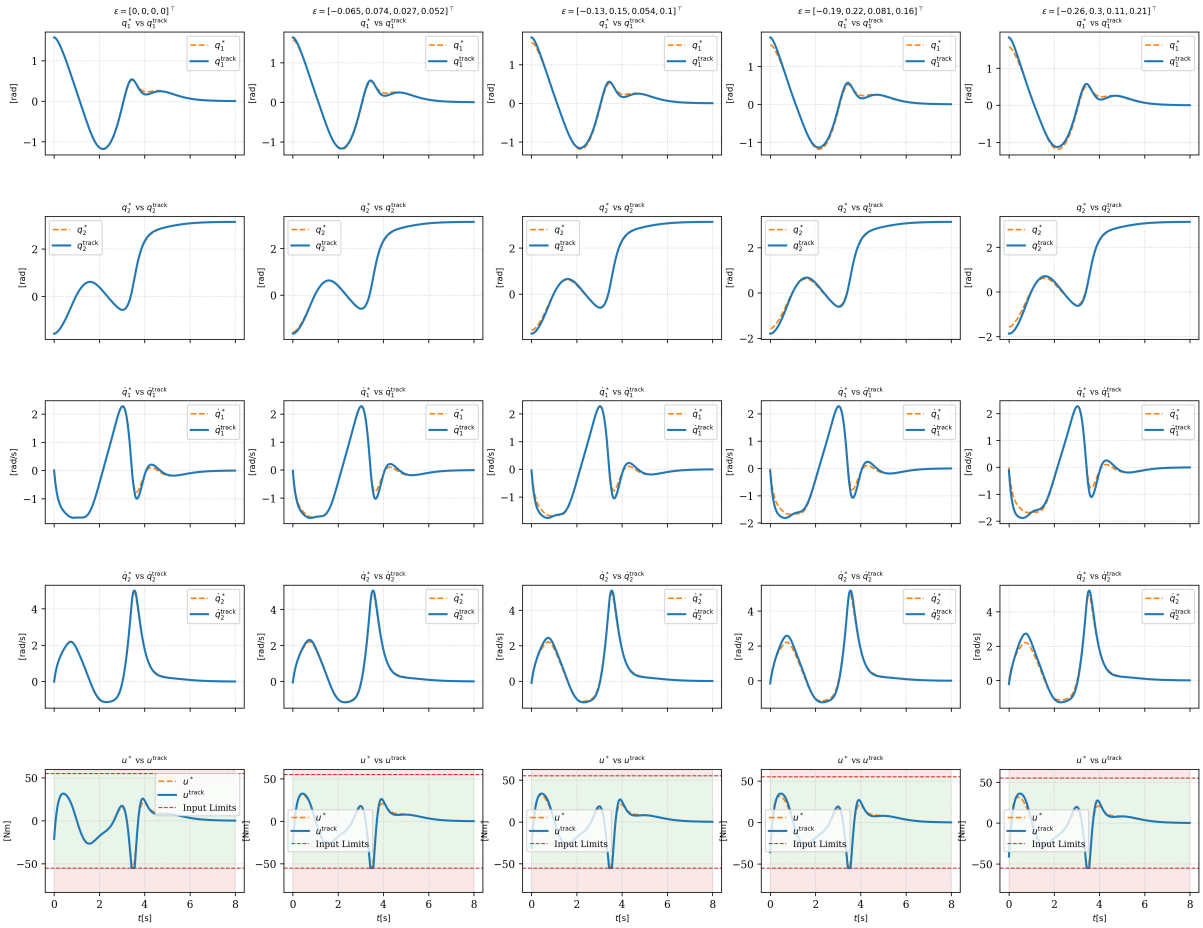


Figure 41: Trajectory tracking on a complex maneuver

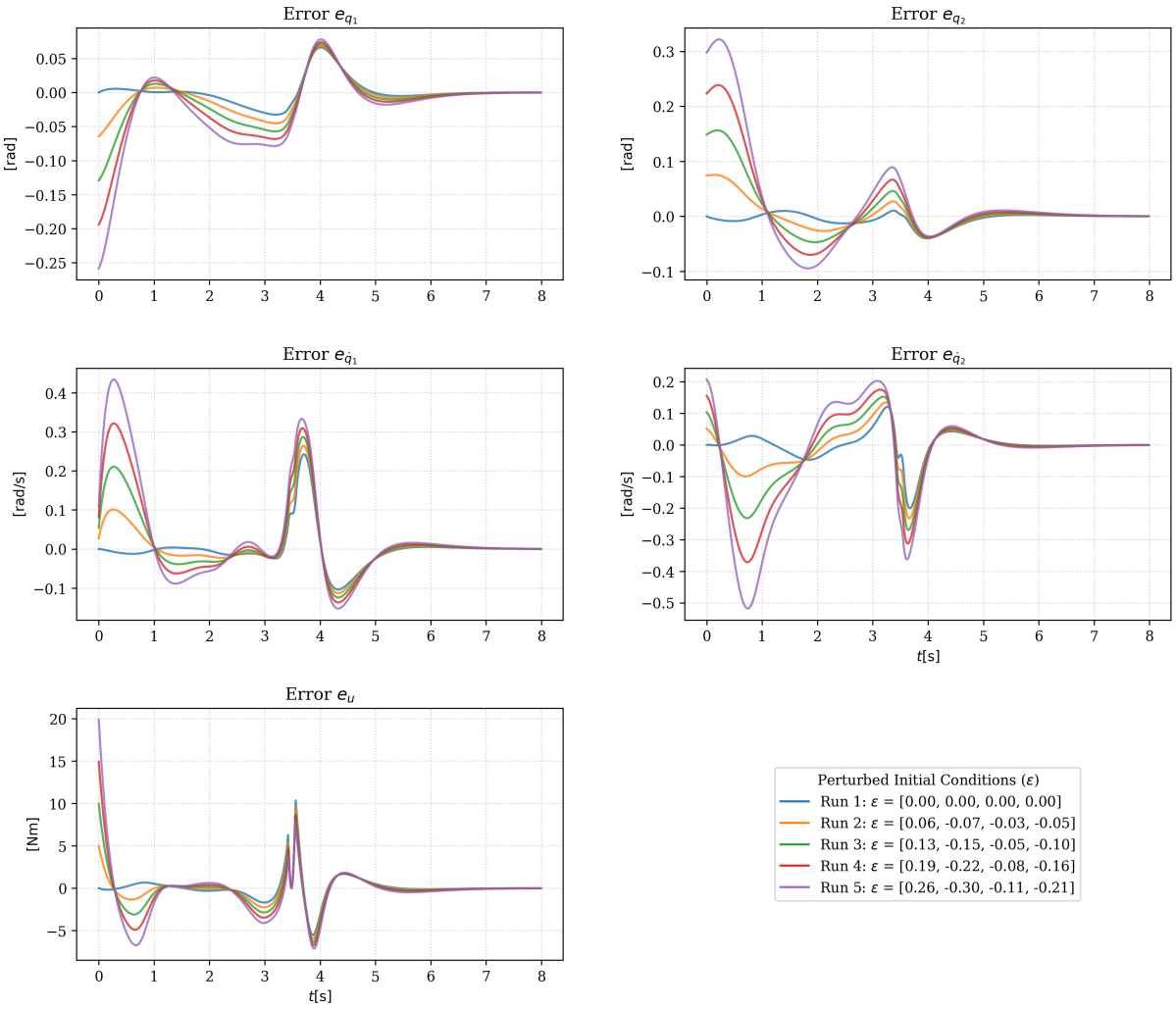
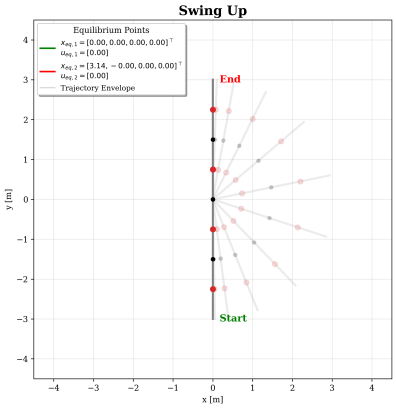


Figure 42: Error analysis during MPC tracking on a complex maneuver

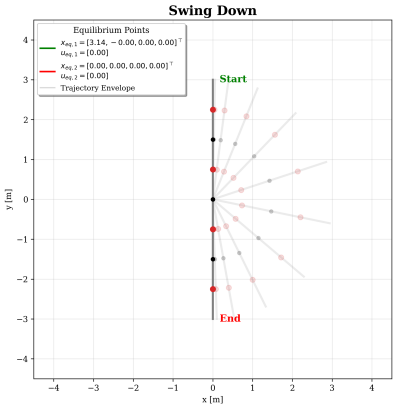
6 Task 5: Animation

6.1 Visualization

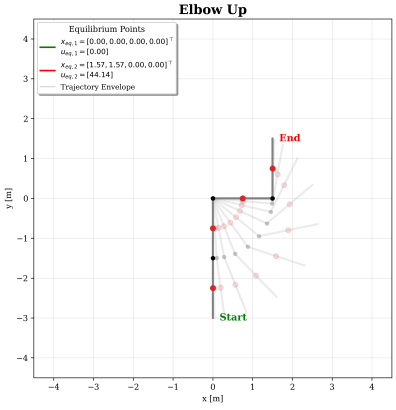
An animation suite was developed `animation.py` to visualize the results. It displays the double pendulum link, the reference trajectory (ghost), and the actual tracked path.



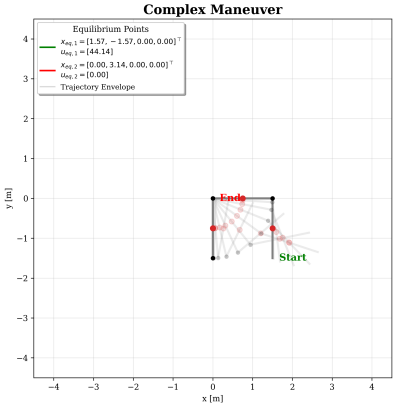
(a) Swing Up



(b) Swing Down



(c) Elbow Up



(d) Complex Maneuver

Figure 43: Double Pendulum Trajectories. Click any image to play the corresponding animation.

7 Conclusion

This work presented a comprehensive framework for the control of a double pendulum with nonlinear joint stiffness, addressing the **challenges of under-actuation and chaotic dynamics**.

Regarding trajectory generation, we showed that standard **Newton-based approaches** are limited in **complex scenarios**. The introduction of the **Dynamic Horizon Extension** proved crucial for generating **complex feasible paths** that not only reach the target but maintain terminal stability.

Finally, we implemented and compared two tracking controllers: a Linear Time-Varying Linear Quadratic Regulator (LTV-LQR), and a Model Predictive Control (MPC). The comparison revealed that **MPC** is essential when operating **near actuator limits**, as it explicitly handles constraints that the LTV-LQR cannot.

On the other hand, the **LTV-LQR** demonstrated superior performance in terms of **computational efficiency and ease of implementation** for less demanding tasks.

In the end, the results were visualized using a custom simulation tool, confirming that the proposed framework can successfully generate and track complex trajectories.

8 Appendix

8.1 Models matrices and parameters

$$M(q) = \begin{bmatrix} I_1 + I_2 + l_{c1}^2 m_1 + m_2(l_1^2 + 2l_1 l_{c2}\cos(\theta_2) + l_{c2}^2) & I_2 + l_{c2} m_2 (l_1 \cos(\theta_2) + l_{c2}) \\ I_2 + l_{c2} m_2 (l_1 \cos(\theta_2) + l_{c2}) & I_2 + l_{c2}^2 m_2 \end{bmatrix}$$

$$C(q, \dot{q}) = \begin{bmatrix} -l_1 l_{c2} m_2 \dot{\theta}_2 \sin(\theta_2) & -l_1 l_{c2} m_2 (\dot{\theta}_1 + \dot{\theta}_2) \sin(\theta_2) \\ l_1 l_{c2} m_2 \dot{\theta}_1 \sin(\theta_2) & 0 \end{bmatrix}$$

$$G(q) = \begin{bmatrix} g l_{c1} m_1 \sin(\theta_1) + g m_2 (l_1 \sin(\theta_1) + l_{c2} \sin(\theta_1 + \theta_2)) \\ g m_2 l_{c2} \sin(\theta_1 + \theta_2) + 3k \sin(\theta_2) \cos^2(\theta_2) \end{bmatrix}$$

$$F = \begin{bmatrix} f_1 & 0 \\ 0 & f_2 \end{bmatrix}$$

Symbol	Description	Value	Unit
m_1, m_2	Mass of link 1 and 2	2	kg
l_1, l_2	Length of link 1 and 2	1.5	m
r_1, r_2	Distance to center of mass	0.75	m
I_1, I_2	Moment of inertia	1.5	$\mathrm{kg\,m^2}$
g	Gravitational acceleration	9.81	$\mathrm{m/s^2}$
f_1, f_2	Viscous friction coefficient	1.0	$\mathrm{N\,m\,s/rad}$
k	Nonlinear angular stiffness	0.6	$\mathrm{N\,m/rad}$

Table 1: Model Parameters (Set 2)

8.2 Analytical Derivation of the Jacobian

This appendix details the analytical computation of the Jacobian matrices A and B used in the linearization of the flexible double pendulum dynamics.

8.2.1 State-Space Formulation

Consider the state space formulation given in Section 1.

$$\dot{x} = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ M(q)^{-1} \left(\tau - C(q, \dot{q})\dot{q} - G(q) - F\dot{q}\right) \end{bmatrix} \tag{12}$$

$$\dot{x} = f(x, u)$$

To linearize the system about a point $(\bar{x}, \bar{u})$, we compute the State and Input Jacobians:

$$A = \left.\frac{\partial f(x,u)}{\partial x}\right|_{\bar{x},\bar{u}} \in \mathbb{R}^{4\times 4} \quad B = \left.\frac{\partial f(x,u)}{\partial u}\right|_{\bar{x},\bar{u}} \in \mathbb{R}^{4\times 1}$$

The state Jacobian $A \in \mathbb{R}^{4\times 4}$ has the following block structure:

$$A = \begin{bmatrix} \frac{\partial \dot{q}}{\partial q} & \frac{\partial \dot{q}}{\partial \dot{q}} \\ \frac{\partial \ddot{q}}{\partial q} & \frac{\partial \ddot{q}}{\partial \dot{q}} \end{bmatrix} = \begin{bmatrix} 0_{2\times 2} & I_{2\times 2} \\ A_{21} & A_{22} \end{bmatrix} \tag{13}$$

8.2.2 Block A_{22}: Partial Derivative w.r.t. Velocities

Let $\eta = \tau - C(q,\dot{q})\dot{q} - G(q) - F\dot{q}$ be the net torque. Then $\ddot{q} = M^{-1}\eta$. Since M and G depend only on positions q, the derivative with respect to $\dot{q}$ is:

$$A_{22} = \frac{\partial \ddot{q}}{\partial \dot{q}} = M^{-1}\left(-\frac{\partial (C(q,\dot{q})\dot{q})}{\partial \dot{q}} - F\right) \tag{14}$$

8.2.3 Block A_{21}: Partial Derivative w.r.t. Positions

The derivative with respect to q involves the chain rule on both the inverse mass matrix and the force vector η:

$$A_{21} = \frac{\partial \ddot{q}}{\partial q} = \underbrace{\frac{\partial (M^{-1})}{\partial q}\eta}_{\text{term 1}} + \underbrace{M^{-1}\frac{\partial \eta}{\partial q}}_{\text{term 2}} \tag{15}$$

Term 1: Derivative of Inverse Mass Matrix Using the identity $\frac{\partial M^{-1}}{\partial \alpha} = -M^{-1}\frac{\partial M}{\partial \alpha}M^{-1}$, the first term is computed column-wise for q_1 and q_2:

$$\frac{\partial (M^{-1})}{\partial q_i}\eta = -M^{-1}\frac{\partial M}{\partial q_i}M^{-1}\eta = -M^{-1}\frac{\partial M}{\partial q_i}\ddot{q} \tag{16}$$

Term 2: Derivative of Forces The derivative of the net torque η with respect to q is:

$$\frac{\partial \eta}{\partial q} = -\frac{\partial (C(q,\dot{q})\dot{q})}{\partial q} - \frac{\partial G}{\partial q} \tag{17}$$

8.2.4 Jacobian Matrix B

The input Jacobian $B \in \mathbb{R}^{4\times 1}$ is straightforward, as only the torque τ depends on u:

$$B = \begin{bmatrix} \frac{\partial \dot{q}}{\partial u} \\ \frac{\partial \ddot{q}}{\partial u} \end{bmatrix} = \begin{bmatrix} 0_{2\times 1} \\ M^{-1}\frac{\partial \tau}{\partial u} \end{bmatrix} \tag{18}$$

Since $\tau = [u, 0]^\top$, $\frac{\partial \tau}{\partial u} = [1, 0]^\top$. Thus, the bottom block of B is simply the first column of M^{-1}.

8.2.5 Explicit Derivative Matrices

For completeness, the non-zero partial derivatives implemented in the code are listed below. Let $s_i = \sin(q_i)$, $c_i = \cos(q_i)$, $s_{ij} = \sin(q_i + q_j)$, $c_{ij} = \cos q_i + q_j$.

Gravity Gradient

$$\frac{\partial G}{\partial q} = \begin{bmatrix} g(m_1 l_{c1} + m_2 l_1)c_1 + m_2 g l_{c2} c_{12} & m_2 g l_{c2} c_{12} \\ m_2 g l_{c2} c_{12} & m_2 g l_{c2} c_{12} + 3k(c_2^2 - 2s_2^2)c_2 \end{bmatrix} \tag{19}$$

Coriolis Velocity Gradient

$$\frac{\partial (C\dot{q})}{\partial \dot{q}} = 2m_2 l_1 l_{c2} s_2 \begin{bmatrix} -\dot{q}_2 & -(\dot{q}_1 + \dot{q}_2) \\ \dot{q}_1 & 0 \end{bmatrix} \tag{20}$$

Mass Matrix Gradient

$$\frac{\partial M}{\partial q_2} = -m_2 l_1 l_{c2} s_2 \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \tag{21}$$

8.3 Derivation of Discrete-Time Jacobians

This appendix details the linearization of the discrete-time dynamics obtained via numerical integration. Let the continuous-time dynamics be $\dot{x} = f(x, u)$, and let the discrete-time system be denoted by:

$$x_{k+1} = F(x_k, u_k) \tag{22}$$

where F is the numerical integration map over a time step Δt. We seek the discrete-time state Jacobian A_d and input Jacobian B_d:

$$A_d = \left.\frac{\partial F}{\partial x}\right|_{x_k,u_k}, \quad B_d = \left.\frac{\partial F}{\partial u}\right|_{x_k,u_k} \tag{23}$$

Let the continuous-time Jacobians evaluated at a specific state x and input u be denoted as $A_c(x, u) = \nabla_x f(x, u)^\top$ and $B_c(x, u) = \nabla_u f(x, u)^\top$.

8.3.1 Forward Euler Discretization

The Forward Euler method approximates the dynamics as:

$$x_{k+1} = x_k + \Delta t f(x_k, u_k) \tag{24}$$

Differentiating with respect to x_k and u_k yields the standard first-order approximation:

$$A_d = I + \Delta t A_c(x_k, u_k) \tag{25}$$
$$B_d = \Delta t B_c(x_k, u_k) \tag{26}$$

8.3.2 Runge-Kutta 4 (RK4) Discretization

The explicit RK4 integration scheme is defined by:

$$x_{k+1} = x_k + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{27}$$

where the intermediate slopes k_i are:

$$\begin{aligned} k_1 &= f(x_k, u_k) \\ k_2 &= f(x_k + 0.5\Delta t k_1, u_k) \\ k_3 &= f(x_k + 0.5\Delta t k_2, u_k) \\ k_4 &= f(x_k + \Delta t k_3, u_k) \end{aligned}$$

To find A_d and B_d, we apply the chain rule recursively to the slopes k_i.

State Jacobian A_d Differentiating the update equation with respect to x_k:

$$A_d = I + \frac{\Delta t}{6}\left(\frac{\partial k_1}{\partial x} + 2\frac{\partial k_2}{\partial x} + 2\frac{\partial k_3}{\partial x} + \frac{\partial k_4}{\partial x}\right) \tag{28}$$

Let $x^{(i)}$ denote the argument of f for the calculation of k_i. The gradients are derived recursively:

$$\frac{\partial k_1}{\partial x} = A_c(x_k, u_k) \tag{29}$$
$$\frac{\partial k_2}{\partial x} = A_c(x^{(2)}, u_k)\left(I + 0.5\Delta t\frac{\partial k_1}{\partial x}\right) \tag{30}$$
$$\frac{\partial k_3}{\partial x} = A_c(x^{(3)}, u_k)\left(I + 0.5\Delta t\frac{\partial k_2}{\partial x}\right) \tag{31}$$
$$\frac{\partial k_4}{\partial x} = A_c(x^{(4)}, u_k)\left(I + \Delta t\frac{\partial k_3}{\partial x}\right) \tag{32}$$

Input Jacobian B_d Differentiating the update equation with respect to u_k:

$$B_d = \frac{\Delta t}{6}\left(\frac{\partial k_1}{\partial u} + 2\frac{\partial k_2}{\partial u} + 2\frac{\partial k_3}{\partial u} + \frac{\partial k_4}{\partial u}\right) \tag{33}$$

Note that u_k appears both directly in f and indirectly through the state perturbation in the intermediate steps. The recursive gradients are:

$$\frac{\partial k_1}{\partial u} = B_c(x_k, u_k) \tag{34}$$

$$\frac{\partial k_2}{\partial u} = A_c(x^{(2)}, u_k)\left(0.5\Delta t\frac{\partial k_1}{\partial u}\right) + B_c(x^{(2)}, u_k) \tag{35}$$

$$\frac{\partial k_3}{\partial u} = A_c(x^{(3)}, u_k)\left(0.5\Delta t\frac{\partial k_2}{\partial u}\right) + B_c(x^{(3)}, u_k) \tag{36}$$

$$\frac{\partial k_4}{\partial u} = A_c(x^{(4)}, u_k)\left(\Delta t\frac{\partial k_3}{\partial u}\right) + B_c(x^{(4)}, u_k) \tag{37}$$

This exact linearization of the RK4 integrator ensures that the discrete-time Jacobians accurately reflect the curvature of the numerical integration path, which is critical for the stability of Newton-based optimal control solvers.

8.4 Newton Method with Dynamic Horizon Extension algorithm

Algorithm 1 Newton Method with Dynamic Horizon Extension

Require: Dynamics $x_{t+1} = f(x_t, u_t)$, Start x_{start}, Goal (x_{eq}, u_{eq}), Costs Q, R, Initial Horizon T, Max Horizon T_{max}, Extension ΔT, Tolerances $\epsilon_{desc}, \epsilon_{term}$.
Ensure: Optimal Trajectory $\mathbf{x}^*, \mathbf{u}^*$

Initialization:
Compute P_∞ solving DARE(A_{eq}, B_{eq}, Q, R) at x_{eq}.
Generate Reference $\mathbf{x}^{ref}, \mathbf{u}^{ref}$ (e.g., Poly5).
Initialize guess $\mathbf{x}^{(0)}, \mathbf{u}^{(0)}$ (Warm start or Equilibrium).
$k \leftarrow 0$ ▷ Optimization Iteration
$j \leftarrow 0$ ▷ Extension Phase
while $T < T_{max}$ **do** ▷ **Outer Loop: Horizon Extension**
  repeat ▷ **Inner Loop: Newton Optimization**
    $J^{(k)} \leftarrow$ Compute Total Cost
    // Backward Pass
    $P_\infty \leftarrow P_f, \quad p_N \leftarrow \nabla_x \ell_f(x_N)$
    for $t = N - 1$ **to** 0 **do**
      Linearize f to get A_t, B_t
      Compute gradients q_t, r_t and Hessian H_t of cost
      $Q_{xx}, Q_{uu}, Q_{xu} \leftarrow$ entry of Value function Hessian
      $K_t \leftarrow -Q_{uu}^{-1} Q_{ux}$ ▷ Feedback Gain
      $\sigma_t \leftarrow -Q_{uu}^{-1} Q_u$ ▷ Feedforward term
      Update P_t, p_t (Riccati equations)
    end for
    // Line Search (Armijo)
    Find α such that cost decreases sufficiently
    // Forward Pass
    $x_0 \leftarrow x_{start}$
    for $t = 0$ **to** $N - 1$ **do**
      $u_t \leftarrow u_t^{(k)} + \alpha\sigma_t + K_t(x_t - x_t^{(k)})$
      $x_{t+1} \leftarrow f(x_t, u_t)$
    end for
    Update $\mathbf{x}^{(k+1)}, \mathbf{u}^{(k+1)}$, increment k
  until $\|\text{Descent}\| < \epsilon_{desc}$
  // Check Terminal Constraint
  if $\|x_N - x_{eq}\| < \epsilon_{term}$ **then**
    Break ▷ Solution found
  else
    Extend Horizon: $T \leftarrow T + \Delta T$
    Pad $\mathbf{x}^{ref}, \mathbf{u}^{ref}$ with x_{eq}, u_{eq}
    Pad current solution $\mathbf{x}, \mathbf{u}$ with simulation at u_{eq}
  end if
end while
return $\mathbf{x}, \mathbf{u}$